

Speaking the Aggressor’s Language: Identity and Sanctions Evasion ^{*}

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Abstract

Export controls work only if third countries decline to circumvent them. We study how language, as part of identity, affects sanctions circumvention in a bystander state in the Russia-Ukraine war. In a difference-in-differences framework for Kazakhstan’s trade, re-exports of sanctioned goods rose far more where Russian was widely spoken among ethnic Kazakhs before 2022. A pre-registered survey experiment identifies the mechanism at the individual level: respondents randomly addressed in Russian on the introductory page are more likely to justify the invasion and help Russia evade sanctions, although the cue leaves their information and incentives unchanged, and the effect is specific to Russia-related attitudes. Finally, we show that the Russian language *polarized* attitudes toward Russia after the invasion and increased online searches for pro-Kremlin news, exposing the boundary that Kazakhstan’s nation-building has yet to close.

Keywords: export sanctions, language, sanctions evasion, Kazakhstan, Russia-Ukraine war

JEL: F51, P45, D74, Z13

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Armed conflict reshapes economic exchange far beyond the battlefield. A growing literature shows that war cuts trade between groups whose identities it sets against one another, even in places where no fighting occurs ([Korovkin and Makarin 2023](#)). Much less is known about an adjacent case, namely countries that are not active parties in the war at all. These bystanders do, however, play an important role in determining the outcome of the economic pressure of modern warfare. Export controls work only if firms in neutral third countries decline to reroute the goods those controls cover, and many do not decline: rerouting through friendly neighbors offsets a large part of the intended decline in the sanctioned country's imports ([Chupilkin et al. 2026](#); [Egorov et al. 2026](#)). Who supplies a sanctioned aggressor, and why, is the question of this paper.

The standard answer is profit. Sanctions make goods scarce in the target country, and circumvention is arbitrage: whoever can collect the rents with minimal risk would do that. We ask whether identity is also part of the answer, i.e., could citizens of a bystander country help a former colonizing power wage war because they have come to identify with it? In an attacked country, war forces a choice; citizens rally around the flag, and the population consolidates around the nation even where prior ties to the aggressor were deepest ([Abramenko et al. 2026](#)). A bystander faces no such imperative. Instead, it inherits the pull toward the former metropole, carried above all by language, which in much of the post-Soviet world still groups citizens into information worlds and identities.

Establishing that this pull is causal is the second gap we address. Language is not randomly assigned. Regions where a former colonizer's language is spoken also lie closer to it, are more likely to share similar ethnic groups, and face lower transaction costs when contracting with its firms, so a correlation between language and trade with the metropole is consistent with identity, geography, and transaction-cost explanations. Isolating the language effect requires exogenous variation, holding information, incentives, and other fixed traits constant. This paper leverages such variation and shows that language moves attitudes toward a foreign power's war and toward evading the sanctions against it.

Kazakhstan after February 2022 is a natural laboratory for this investigation. First, the country is not at war but shares a border, a customs union, and a colonial past with Russia, so we observe how citizens respond when war makes an identity salient without the destruction

and mobilization that confound studies of countries at war. Second, more than thirty years after independence, and although the state language is Kazakh, Russian remains the everyday language of a large share of the population, including many ethnic Kazakhs. Speaking Russian and being ethnic Russian therefore come apart, enabling identification of the standalone effect of language as a part of identity. Third, every region faced the same post-invasion Russian demand, the same customs union, and the same profit opportunity, so uptake can be compared across regions while holding incentives fixed. Finally, Kazakhstan became a re-exporter of sanctioned goods reaching Russia in defiance of Western export controls, and its unique transaction trade records make that sanctions circumvention observable at the sub-national level.

We bring four data sources. The first is the universe of Kazakhstan’s trade transactions with the Eurasian Economic Union (EAEU) from January 2021 through September 2023, aggregated to the region, month, product-class, and destination level and merged with the regional share of Russian-speaking ethnic Kazakhs from the 2021 Census, which we call Russian-language penetration. The second is an original, pre-registered online survey experiment of 5,025 respondents fielded in November 2023, which randomizes the language of the introductory page before any question is asked. The third is eight waves of the Central Asia Barometer (CAB) spanning 2020 to 2023, four fielded before the full-scale invasion and four after it. We finally present regional online search data on news sources.

Our main result is that Kazakhstan’s exports of sanctioned goods to Russia surged after the invasion, and most where Russian is widely spoken among ethnic Kazakhs. Before the invasion, exports of the goods that Western countries would later sanction – electronics and machines – were small and flat, at 2.4 million USD per region-month against 8.4 million for all other goods. After February 2022, they rose by 9.8 million USD per region-month, more than quadrupling, while exports of the same product classes to the other EAEU members rose by only 0.6 million. Domestic production of these goods did not move, which identifies the flows as re-exports. We then estimate a triple difference-in-differences model in which treatment exposure is the regional share of Russian-speaking ethnic Kazakhs from the pre-invasion Census, and we absorb post-invasion changes in border regions through a border-by-post interaction. The invasion by itself does not explain these exports: extrapolated to a

region with no Russian-speaking Kazakhs, the estimated effect is indistinguishable from zero. Re-export growth follows the language gradient, and it is about three times larger in the most than in the least Russian-speaking region. Moving from the least to the most Russian-speaking region, from 27 to 66 percent, raises export growth of sanctioned goods to Russia by a further 10.3 million USD per region and month.

Even though the parallel trends assumption holds, trade data alone cannot separate the effect of Russian language from other factors that correlate with regional language patterns pre-war and affect sanctions circumvention since 2022.¹ Our survey experiment provides that separation. Respondents who clicked our online ad were shown a consent page in a randomly assigned language, Kazakh or Russian, and could switch languages at any time. The cue works as a first stage on identity itself: being addressed in Russian raises the probability of completing the entire survey in Russian by 22 percentage points, a revealed choice rather than a stated one, and it also raises reported use of Russian with friends and for news, though not at home, where language is most habitual. The cue then moves attitudes. Respondents assigned the Russian introductory page are 3 percentage points more likely to consider helping Russia evade Western sanctions justified, 4 percentage points more likely to consider the invasion justified, and hold a more favorable opinion of Russia itself. Instrumenting the endogenous survey language with the randomly assigned language, we show that the Russian language raises justification of sanctions evasion by 15 percentage points and of the invasion by 18 percentage points. Crucially, the effect is specific to Russia-related affairs: the same cue does not affect opinions of China and the United States, trust in the national government, the justifiability of protest participation, and the justifiability of overworking.

Two further pieces of evidence reveal the *polarization* pattern induced by the Russian-language penetration after February 2022 and explain why it is sustained. For the eight CAB repeated cross-section waves covering 2020-2023, using difference-in-differences, we show that respondents from regions with more Russian-speaking Kazakhs were, if anything, slightly *less* favorable toward Russia before the invasion. The positive effect of the Russian-language penetration appears in the first wave after February 2022 and persists through late 2023

¹Our additional specification shows that the main result holds when we extend the main specification with the ethnicity-by-post interactions. Nevertheless, the Russian language can also reduce the transaction cost of trade with Russia.

even though we control for individual characteristics including ethnicity. Nothing comparable happens for opinions of the United States or China. Yandex online search data then show that Russian-language penetration increased pro-Kremlin news search: moving from the least to the most Russian-speaking region raises searches for war-related keywords by 13.3 per 1,000 inhabitants, with comparable increases for searches of pro-Kremlin aggregators and outlets, and only a small increase in searches for Western outlets.

We consider and rule out the main alternative explanations. In the trade results, proximity and ethnicity are absorbed by the border-by-post and ethnicity-by-post interactions. Russian as a less costly contracting language predicts the same trade pattern after February 2022 but not the rest of the evidence: Russia-related attitudes should not respond to a randomized introductory language, yet they do; and language should not polarize attitudes toward Russia, while it does.

Related literature and contribution. Our paper makes four contributions. First, we contribute to the economics of identity. Since [Akerlof and Kranton \(2000\)](#) placed group identity in the utility function, this literature has treated identity as a choice: people weigh a group’s status against their perceived distance from it ([Shayo 2009, 2020](#); [Gidron and Shayo 2026](#)), and they respond to incentives by reclassifying their race ([Dahis et al. 2019](#)), choosing their children’s ethnicity ([Jia and Persson 2021](#)), or expressing identity through consumption ([Atkin et al. 2021](#)). We show how a randomized cue shifts identity choice in a setting where it carries material consequences, and contribute a design – randomizing the language in which respondents are addressed, before any question is asked – that is cheap and portable to any multilingual setting, adding to the measurement toolkit surveyed by [Rohner and Zhuravskaya \(2025\)](#). We disentangle ethnicity from language and thus add to the discussion around identity traits of culture ([Desmet et al. 2017](#)) and language ([Pérez and Tavits 2017, 2019](#); [Marquardt 2022](#)). In particular, we expand the literature on Russia’s influence over the post-Soviet space, which attributes loyalty to Russian ethnicity ([Peisakhin and Rozenas 2018](#); [Korovkin and Makarin 2023](#); [Larreguy and Martinez 2024](#)).²

²On the non-causal effect and persistence of imperial languages, see [Phillipson \(2008\)](#) and [Crystal \(2012\)](#) on the British empire, [Neathery-Castro and Rousseau \(2005\)](#) and [Vigouroux \(2013\)](#) on the French, and [Mar-Molinero \(2000\)](#) and [Stavans \(2021\)](#) on the Spanish.

Second, we contribute to the literature on nation-building and on how wars redraw group boundaries (Rohner and Zhuravskaya 2023, 2025). Shared exposure to threat or repression forges a common identity in the population (Dehdari and Gehring 2022; Gehring 2022). Ukraine consolidated after the full-scale Russian invasion, which made the boundary between the two states the salient divide and raised the status of the Ukrainian nation, and identification shifted toward the nation most strongly in the districts with the most ethnic Russians (Abramenko et al. 2026). We show that Kazakhstan instead polarized: the war raised the salience of a trait its residents sit at opposite ends of (language) and pulled them apart. Consolidation and polarization are thus different outcomes of the same mechanism under different patterns of exposure. The direction depends on which trait a war makes salient. In other bystanders of Russia’s 2014 invasion – Estonia and Latvia – the majority identified more strongly with the European Union while the Russian-speaking minority did not (Gehring 2022). In Kazakhstan, we identify an event and the mechanism driving the population apart. This also connects to work on identities formed by dividing events such as Brexit (Hobolt and Tilley 2026) or vaccination status (Bor et al. 2023; Henkel et al. 2023; Wagner and Eberl 2024).

Third, we connect identity to international trade and to the effectiveness of sanctions (Drezner 2024). A shared language with a former colonial power is usually an asset, lowering the cost of communication and contracting and raising trade (Melitz 2008; Melitz and Toubal 2014; Guiso and Makarin 2020; Gurevich et al. 2025). We show that for conflict bystanders and under sanctions, economic ties among groups sharing the same identity can be intensified. This provides a micro-level foundation for the role of Central Asia in Russia’s evasion of Western sanctions (Chupilkin et al. 2026; Babina et al. 2023; Egorov et al. 2025, 2026), in which circumvention is modelled as arbitrage. We instead show that the rents were collected mostly where language ties people to Russia. Forgoing profit, rather than taking it, is what makes identity economically essential (Atkin et al. 2021).

Fourth, we speak to the literature on media as an instrument of cross-border influence. Where the state controls the media environment, framing shifts attitudes regardless of predispositions (Enikolopov et al. 2011; Zhuravskaya et al. 2020; Pan et al. 2022). Where audiences can choose, effects run along prior loyalties: Russian TV raised pro-Russian voting in Ukraine among viewers already predisposed toward Russia while backfiring among the rest (Peisakhin

and Rozenas 2018), independent television inside Russia polarized rather than persuaded (Enikolopov et al. 2026), and Beijing-backed media in Taiwan did the same (Kao 2026). This literature studies what media do to a given audience. We show how the audience forms in the first place. Media consumption is a choice based on identities, such that media consumption amplifies and stabilizes an identity that was determined earlier.

The remainder of the paper proceeds as follows. Section 1 provides background and context, and Section 2 develops the conceptual framework and derives its predictions. Section 3 describes the data. Section 4 presents our main trade results. Section 5 turns to the mechanism: Section 5.1 reports the survey experiment, Section 5.2 traces the dynamics of attitudes in the CAB waves, Section 5.3 examines online search behavior, and Section 5.4 rules out alternative explanations. Section 6 concludes.

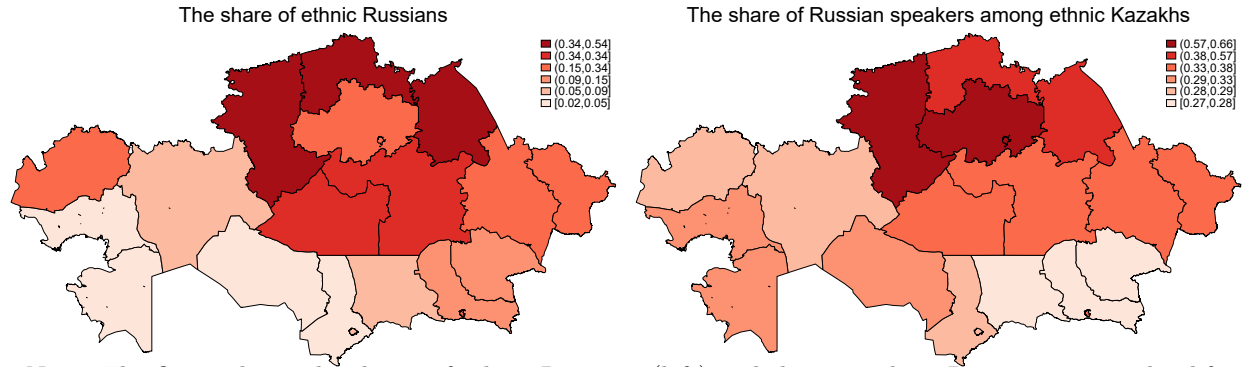
1 Background and Context

One of the successor states of the Golden Horde, the Kazakh Khanate was gradually conquered and absorbed by the Russian Empire from the late 18th to the mid-19th century. Following the Russian Revolution, the territory became part of the Soviet Union. After several territorial reorganizations, Kazakhstan’s modern borders were established in 1936 with the formation of the Kazakh Soviet Socialist Republic. Kazakhstan declared its independence from the Soviet Union on December 16, 1991 and has since been an independent state. Kazakhstan, like other former colonies, has seen real decolonization efforts, though its stance toward Moscow has long remained ambivalent.

As a result of two centuries of Russian colonization, a significant Russian minority continues to live in Kazakhstan. Although Kazakh is the official state language, Russian also holds the status of an official language, and a substantial portion of the population continues to speak Russian daily. According to the 2021 Census, the share of ethnic Kazakhs is 70%, and that of ethnic Russians is 15.5%. Moreover, among ethnic Kazakhs, the share of people speaking and reading Russian is 79.5%, and the share using Russian in everyday life is 36.5%.³ Figure 1 illustrates the distribution of ethnic Russians and Russian speakers among Kazakhs by region, showing that Russian speakers are concentrated in the north of the coun-

³Results of the National Population Census 2021

Figure 1: Regional heterogeneity by ethnicity and language use



Note. The figure shows the shares of ethnic Russians (left) and those speaking Russian in everyday life among ethnic Kazakhs (right) by regions of Kazakhstan according to the 2021 Census. The Pearson correlation between the regional share of ethnic Russians and the share of Russian speakers among Kazakhs is 0.64.

try and in the two main cities – Astana and Almaty. We show throughout the paper that this regional variation in language use determines political opinions and economic behavior, including trading sanctioned products with Russia.

Since the start of Russia’s war in Ukraine in 2014, the Kazakh government has been careful to maintain a low profile with respect to the conflict, avoiding too extensive coverage of the conflict in the official media and employing “strategic silence” ([Dadabaev and Sonoda 2023](#)), in order to antagonize neither Russia nor the West, even though at times concerns and criticism about the war are voiced.⁴ As a result, almost all of the Kazakh- and Russian-language media published within Kazakhstan take either a neutral or a moderately critical stance towards Russia when covering the conflict.⁵ However, at the same time, Russian-language media from Russia are widely available in Kazakhstan. Especially when searching for information on the internet, the probability that Russian speakers will end up on a site from Russia featuring pro-Kremlin content is high, while Kazakh speakers are more likely to find content that is more neutral with respect to Russia’s invasion of Ukraine.

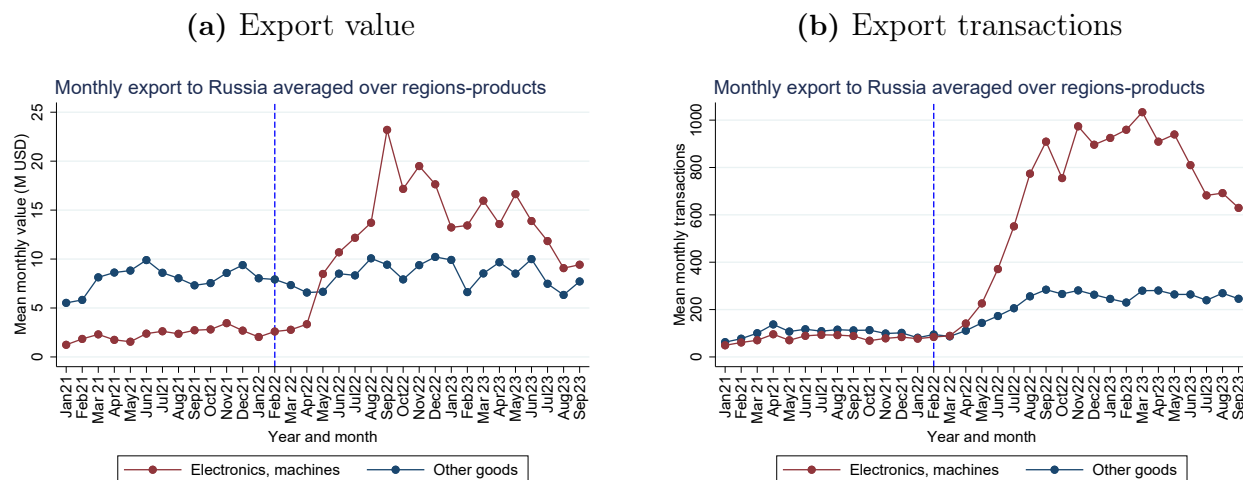
While the Kazakh government has been careful to maintain a neutral stance towards the conflict in its official statements, economically, Kazakhstan has taken a pro-Russian stance.

⁴Steppe change: How Russia’s war on Ukraine is reshaping Kazakhstan. European Council on Foreign Relations, 2023.

⁵This is also a result of Kazakh media being largely state-controlled. According to [Freedom House](#), “the dominant media outlets in Kazakhstan are either in state hands or owned by government-friendly businessmen”.

Exports from Kazakhstan to Russia have increased significantly since February 2022. However, as illustrated by Figure 2, this is *only* the case for electronic goods and machines, i.e., products on which Western countries imposed sanctions after Russia’s full-scale invasion of Ukraine in February 2022, while exports of other goods have remained remarkably stable.⁶ After February 2022, Kazakhstan reduced imports of sanctioned products from Russia (Figure A2), slightly increased their imports from other EAEU countries (Figure A3), and substantially increased imports of sanctioned products from the rest of the world (Figure A4), without significant changes in domestic production (Figure A5). Such cases of sanctions circumvention were also revealed by investigative journalists,⁷ and US Secretary of State Antony Blinken visited Kazakhstan in February 2023 to discuss specifically the issue of sanctions circumvention.

Figure 2: Export from Kazakhstan to Russia



Note. The figure shows the monthly value of export in millions of USD (left) and the number of export transactions (right) of Kazakhstan to Russia, averaged over 5 product classes and regions for 2021-2023. For more details, see Section 3 for the international trade data.

⁶Figure A1 of Appendix A shows Kazakhstan’s export value and transactions to other EAEU countries, excluding Russia. While there is a surge after February 2022, it is of a smaller order.

⁷See, e.g., [Kazakhstan Has Become a Pathway for the Supply of Russia’s War Machine. Here’s How It Works](#). Organized Crime and Corruption Reporting Project, 2023. or [Now through Kazakhstan: devices from the United States, Germany and France, which can be used for electronic warfare, continue to go to Russia](#). The Insider, 2024. or [Shooting Ukraine in the back: Sniper rifles and ammunition from the EU and U.S. are being supplied to Russia despite sanctions](#). The Insider, 2024. or [Russian businessman extradited to Estonia for selling \\$50 million worth of foreign equipment to Russian defense enterprises](#). The Insider, 2025.

2 Conceptual Framework

We start with a conceptual framework that predicts behavior based on the identity choice. We adopt the social identity model of [Shayo \(2009, 2020\)](#) in the form that [Abramenko et al. \(2026\)](#) develop for wartime Ukraine. We also inherit their notation and definitions while adjusting the interpretation for a bystander country – like Kazakhstan. A resident i chooses which of two identities to identify with, $j \in \{N, R\}$, the Kazakh nation N or the transnational Russian-speaking world R . Identification is endogenous and may be linked to the individual’s legacy and current circumstances. The value of identifying with group j is

$$I_{ij} = (1 - \beta) S_j - \beta d_{ij}, \quad (1)$$

where S_j is the group’s *status*, the esteem it enjoys in the wider social order, d_{ij} is i ’s perceived *distance* from the group’s typical member, and $\beta \in (0, 1)$ sets the trade-off between the two. People value belonging to a group more when the group is well regarded and when its members are similar to themselves. Distance aggregates over three traits, $k \in \{\text{language, ethnicity, culture}\}$,

$$d_{ij} = \sum_k w_k \delta_k(i, j), \quad (2)$$

where $\delta_k(i, j)$ is i ’s gap from group j ’s prototype on trait k and $w_k \geq 0$ is an *attention weight* measuring how much trait k is currently on people’s minds. These weights are the active part. An event can thrust a trait into focus and raise its weight without moving anyone’s actual position on it, as when an ethnic war raises the weight on ethnicity ([Sambanis and Shayo 2013](#)). Identity is thus governed by three forces, namely group status, trait salience (weights), and the person’s own position in trait space. Appendix [A](#) gives a numerical example.

[Abramenko et al. \(2026\)](#) apply this framework to Ukraine, where the 2022 invasion moved status and salience in the same direction. Successful resistance and international solidarity raised S_N while military setbacks and isolation lowered S_R . As the attack struck the country as a whole and explicitly denied that a Ukrainian nation exists, the trait thrust into focus was the side of the interstate boundary, a trait every resident shares. Raising the weight on a

shared trait lowers d_{iN} and raises d_{iR} for everyone at once, most sharply for Russian speakers. The framework predicts *consolidation* and the authors find districts with more ethnic Russians converging toward the national mean and residents switching to Ukrainian in language, music, and baby names. Our contribution is to apply the framework to a *bystander* country whose citizens have the identity R available to them. A bystander lacks precisely the forces that consolidated Ukraine. No resistance lifts S_N , whatever the war does to S_R is common to every resident and cannot pull two groups in opposite directions, and ethnic distances are large and fixed, as ethnic Kazakhs identify as Asian rather than Slavic. What the war moves is the salience of language, because Russia declared the invasion a war for the “Russian world” and the protection of Russian speakers wherever they live.⁸ Language, unlike the interstate boundary in Ukraine, is a trait on which residents sit at opposite ends, so the higher weight w_{lang} magnifies gaps that may already exist, making R better to identify with for a Russian speaker and worse for a Kazakh speaker. Where the attacked country consolidates around a trait its residents share, the framework predicts that a bystander *polarizes* along the trait that divides its residents, with conformity following downstream as those newly identifying with R consume Russian-language media, defend Russia’s war, and sustain the trade that supplies it. This logic yields five predictions that we take to the data.

Prediction 1 (Status). After February 2022, the average opinion of Russia should fall, because the war lowers Russia’s status S_R and this component of equation (1) enters identically for every resident. Being common to all, it shifts the level of pro-Russian sentiment without distinguishing between language groups. The fall should be specific to Russia, since the shock is to S_R and not to how residents regard third countries such as China or the United States.

Prediction 2 (Polarization). After February 2022, an attitude gap between Russian and Kazakh speakers should widen, on top of the common decline in Prediction 1. By polarization we mean a widening gap in attitudes toward Russia between the two language groups, not animosity between the groups themselves. Before the war, language carried little attention weight, so it should predict attitudes only modestly. After the invasion, because the operative trait is language and not ethnicity, the gap should appear even among Russian-speaking ethnic

⁸Abramenko et al. (2026) note that in principle their mechanism “could instead have sharpened internal ethnolinguistic divisions under different patterns of exposure.” The bystander, exposed to the war’s framing but not to its violence, is that case.

Kazakhs, a group that any ethnicity-based account would place firmly with N .

Prediction 3 (Identity activation). The mechanism runs through perceived distance, so an exogenous cue that lowers the felt distance d_{iR} should shift attitudes toward Russia’s positions even without any new information about the war. The shift should be specific, appearing only on topics that map onto the divide between N and R .

Prediction 4 (Behavioral conformity). Identity choice is consequential, so it should surface in survey responses and behavior. The most Russian-speaking regions should sustain trade with Russia after the invasion, including in sanctioned goods, since supplying the group one identifies with is conformity to its interests. The sanctions themselves create scarcity rents in Russia that raise the material return to this trade for every region alike. What the model adds is that uptake of this common profit opportunity varies with identification, concentrated where the Russian-language tie is strong and absent where it is not. Identification thus predicts who collects the rents, which Section 4 tests through the regional language gradient.

Prediction 5 (Reinforcement and persistence). Demand for Russian-language media should rise as an expression of conformity during high-salience moments. Consuming that media in turn lowers d_{iR} , by shifting positions on cultural traits toward the Russian prototype, and raises perceived S_R through propaganda.

3 Data

To understand how Russian-speaking identity affects attitudes and trade patterns toward Russia, we combine several data sources. First, we use transaction-level trade data between Kazakhstan and the Eurasian Economic Union (EAEU) for the period 2021 to 2023 and merge it with the regional share of Russian-speaking Kazakhs from the 2021 Census. It enables us to investigate how Russian-language penetration affects the re-export of sanctioned products to Russia after February 2022. Second, we conducted an original online survey in November 2023 that embeds a pre-registered language-randomization experiment, allowing us to test whether Russian-speaking identity causally shifts attitudes toward the war and sanctions circumvention. Third, using eight waves of a nationally representative telephone survey conducted by the Central Asia Barometer in Kazakhstan in 2020–2023, we test whether pre-war Russian-language penetration polarizes attitudes toward Russia in Kazakhstan. Finally, using data on

online searches on Yandex, we test whether this divergence in political views can be partially explained by differences in online news search after February 2022.

3.1 International trade

We use data from international trade between Kazakhstan and Russia to study the effect of Russian-language penetration on re-export activity. Our dataset was collected from the Bureau of National Statistics of Kazakhstan (www.stat.gov.kz). It consists of all transactions between Kazakhstan and the countries of the Eurasian Economic Union for the period January 2021 – September 2023. Each transaction contains information on the traded product classification code (TNVED, coinciding with the Harmonized System (HS) codes), the trading firm’s region in Kazakhstan, the trading partner’s country, the declared product value (in USD and KZT), and the gross weight. Focusing on export activity from Kazakhstan, comprising 811,000 individual transactions, we classify each of them into five product classes: (i) electronics and machines, (ii) food, (iii) textiles, (iv) metals, and (v) others.⁹ Most of the products from class (i) are under Western sanctions, prohibiting their re-export from Kazakhstan to Russia. We therefore call them *sanctioned products*, even when discussing the period before 2022. Among all export transactions, 78.8% are to Russia and 17.4% to Kyrgyzstan. We aggregate the export data, calculating the export value (in millions of USD) and the number of transactions at the region (17 regions),¹⁰ month (33 months), product class (5 classes), and destination (Russia vs. other EAEU countries) level. Figure 2 shows the dynamics of exports from Kazakhstan to Russia averaged over regions and product classes for electronics/machines and other products (product classes (ii)-(v)). Figure A1 of Appendix A shows these dynamics for exports to other EAEU countries. Finally, the Bureau of National Statistics of Kazakhstan also provides industrial production output data for each region and month for 36 product categories of the national classifier of economic activities (OKED), which we aggregated into the same 5 product classes.¹¹ Figure A5 of Appendix A shows the dynamics of

⁹We allocate the 2-digit TNVED product classifications as follows: electronics (85,88,90,91), machines (84,86,87,89), food (01-24), textiles (41-43, 50-67), metals (72-83), with “others” covering the rest.

¹⁰In 2021, there were 17 regions; in 2022, three regions were split. As we merge the trade data with the 2021 Census data, all export transactions from new regions in 2022 are assigned to their 2021 parent regions.

¹¹We allocate the 2-digit OKED product classifications as follows: electronics (26,27), machines (28,29,30), food (10,11,12), textiles (13,14,15), metals (24,25), with “others” covering the rest.

Kazakhstan’s production averaged over regions and product classes for electronics/machines and other products. Notably, there is no significant change in sanctioned product production.

3.2 Language experiment in an online survey

We have conducted a pre-registered¹² online survey experiment fielded in November 2023 via NAC Analytica, a leading public opinion pollster in Kazakhstan.¹³ Participants were recruited through advertisements on Instagram and compensated 700 Tenge (approx. 1.50 USD).¹⁴

People who clicked the survey ad on Instagram were directed to a survey platform, where the introductory page with the consent form was displayed in a *randomly assigned* language, Kazakh or Russian. This randomization is the centerpiece of our design. Respondents were free to switch language at any point of the survey, including the introductory page. In practice, those who switched did so immediately on the introductory page or the first demographic question, and completed the survey in their chosen language.¹⁵ Table C2 of Appendix C confirms that the randomization was successful: across all demographic characteristics, Cohen’s d is below 0.1.

We have several, potentially endogenous, measures of Russian-speaking identity. The first is the language in which a respondent *completes* the survey, which we consider a *revealed* preference for a language (Atkin et al. 2021). The second set of measures includes *stated* language use in different conditions. We ask which language respondents use most often (i) at home, (ii) with friends or colleagues, and (iii) for news about political and economic events abroad.¹⁶

Our outcomes fall into two groups. The first includes *Russian affairs*: (i) whether respon-

¹²https://aspredicted.org/YPM_LSH

¹³<https://nacanalytica.com/en/>

¹⁴Quality verification included an attention check on respondents’ ages, a minimum response time, and a restriction on duplicate participation by IP address and phone number. Out of 28,021 users who clicked the survey ad on social media to read the consent form, 5,025 completed the survey and passed all quality checks as unique respondents; most who dropped out did so on the consent page or during the initial demographic questions. Attrition is not differential by treatment. The completion rate is 18.1% among users shown the Kazakh introductory page and 17.8% among those shown the Russian one ($p = 0.54$). Among users eligible by age, region, and IP address, the rates are 23.5% and 24.0% ($p = 0.34$). Table C2 of Appendix C reports both comparisons.

¹⁵We tracked the language for each question.

¹⁶We also asked about the set of languages respondents know well.

dents consider it justified to help Russia evade Western sanctions,¹⁷ (ii) whether they consider Russia’s invasion of Ukraine – its “Special Military Operation” (SMO) – justified,¹⁸ and (iii) the overall opinion of Russia. The second group of outcome variables includes *non-Russian affairs*: (iv) opinions of China and the United States, (v) trust in government, (vi) the justifiability of participating in protests for political change, and (vii) whether working more than 50 hours a week is justified. All questions offered a *Prefer not to answer* option.¹⁹ Table C1 reports summary statistics for the survey sample. The share of respondents who favor sanctions circumvention is 30%, and the share who justify the invasion of Ukraine is 26%. The latter is similar to the share justifying the invasion from the computer-assisted telephone interview (CATI) survey of the Central Asia Barometer, which we present next.

3.3 Central Asia Barometer

Since June 2017, the Central Asia Barometer (CAB) has been conducting public opinion surveys in the four Central Asian republics of Kazakhstan, Kyrgyzstan, Uzbekistan, and Tajikistan twice a year in April-June and September-November. For our study, we use waves of the 2020–2023 CAB for Kazakhstan.²⁰ All eight survey waves include questions about attitudes toward Russia, the United States, and China²¹ and a block of demographic questions including ethnicity, language spoken at home, and the language of the interview. Around 15% of respondents declared their primary occupation to be “Business owner/Entrepreneur”, “Director/Head of company or organization” or “Head of department/Middle manager”. We

¹⁷“In your opinion, is helping Russia avoid Western sanctions generally justified or not justified?”

¹⁸“In your opinion, is (Russia’s Special Military Operation in Ukraine/ invasion of Ukraine) generally justified or not justified?”. The wording “Special Military Operation in Ukraine” versus “invasion of Ukraine” was itself randomized to balance framing effects.

¹⁹The survey also fielded a set of list experiments for (i), (ii), (vi), (vii). The placebo list (vii) showed a significant design effect, so we rely on the direct questions and do not analyze the lists in this paper.

²⁰Since 2020, these have been CATI surveys with randomly generated mobile phone numbers drawn from the mobile operators’ number ranges in the respective countries. The last available wave is in Fall 2023. For more details, see <https://www.ca-barometer.org/en>.

²¹The question is: “Thinking about other countries, please tell me if you have a very favorable (4), somewhat favorable (3), somewhat unfavorable (2), or very unfavorable (1) opinion of (Russia/the United States/China).” The waves starting from 2022 also include several questions related to Russia’s full-scale invasion of Ukraine, of which we use the following two in the Appendix: (i) “In your view, who is mainly responsible for the situation in Ukraine?”, and (ii) “To what extent do you think Russia’s special military operation in Ukraine is justified or unjustified?”.

call these respondents “Firm managers”.

Table 1: Descriptive statistics for the language, ethnicity, and opinion of countries

Year	Obs.	Home lang.	Russian (%)		Favorable opinion of (%)		
			Survey lang.	Ethnicity	Russia	US	China
2020	3,500	49.9	85	25	77.1	46.5	46.5
2021	3,500	47.3	78.5	22.2	70.5	51.9	44.9
2022	3,020	41.1	70.3	17.9	50.9	49.4	51.5
2023	3,000	45.3	70.6	20.5	57.1	51.4	60.3

Note. The table uses CAB survey waves for 2020-2023. The “Favorable opinion” columns show the percentage of “Very favorable” or “Somewhat favorable” opinions. Table A1 shows descriptive statistics for the war attitudes and responsibility attribution.

Table 1 presents descriptive statistics for language, ethnicity, and attitudes by year. The share of respondents who speak Russian at home ranged from 41% to 50% and is more than twice the share of ethnic Russians. Notably, the share of respondents completing the survey in Russian dropped from 85% in 2020 to 70.6% in 2023. Holding a favorable opinion of Russia fell from 77% in 2020 to 57% in 2023, with a minimum in 2022. Favorable views of the US have remained remarkably stable, while views of China have improved since 2022.

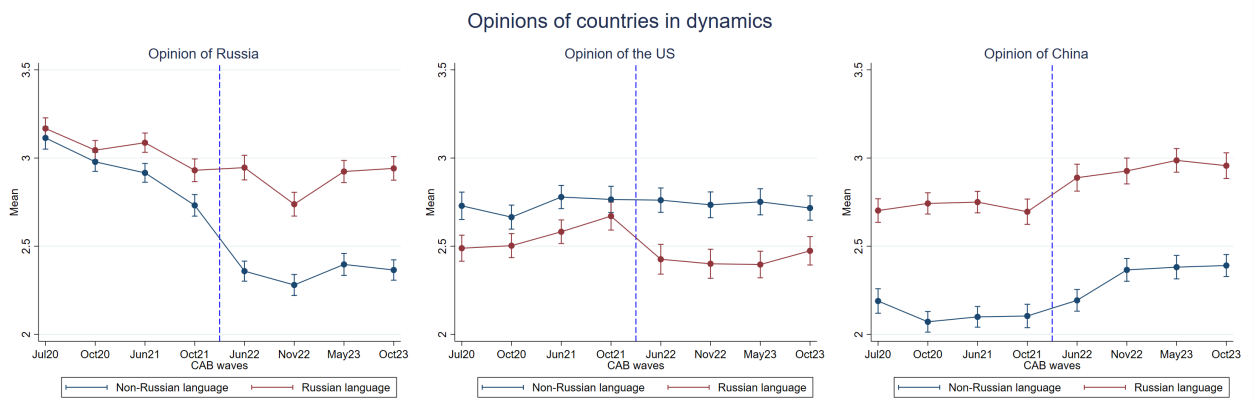
Figure 3 shows the dynamics of opinions toward Russia, the US, and China, split by home language. There was a broadly similar downward trend in opinions toward Russia, with a minor language gap before 2022; the gap widened substantially after 2022. Similarly, US opinion was relatively stable with a minor language gap before 2022; the gap widened substantially thereafter. Opinion of China has evolved in parallel for the two languages before and after 2022. Figure A6 shows similar dynamics split by the survey language.

3.4 Yandex Search Data

We use Yandex search data to study how regional Russian-language penetration affects online search behavior for information about the war.²² The data is taken from wordstat.yandex.kz.

²²Yandex has a [market share](#) of roughly 15% in Kazakhstan, whereas Google handles the vast majority of searches. However, Google search data is not accessible in a systematic and reliable manner due to several limitations. Google Trends relies on a subsampling methodology, meaning the provided statistic is based only on a subset of searches, making the search statistics inconsistent over time. Additionally, Google Trends

Figure 3: Dynamics of Opinions of Russia, the US, and China by home language



Note. The figure shows the mean trends in opinions across different countries by **home** language (Russian or Kazakh), based on CAB data for 2020-2023. Figure A6 shows the dynamics, split by the **survey** language.

It shows the overall number and percentage of searches for a specific phrase per month and region. Figure 4 shows a substantial spike in searches for the phrase “war in Ukraine” in Russian in February 2022, with a smaller spike around the Russian mobilization in September 2022 and a stabilization after that.²³ Figures A9, A10, A11, and A12 in Appendix A document that interest in internet news more broadly also spiked in February 2022. In addition, we document that while searches for information from pro-Kremlin news aggregators Yandex Dzen and Yandex News (Figures A10 and A11 of Appendix A) were significant and growing after February 2022, there is only a minor increase in searches for Euronews, the most popular pro-Western news channel in Kazakhstan (Figure A12 of Appendix A).

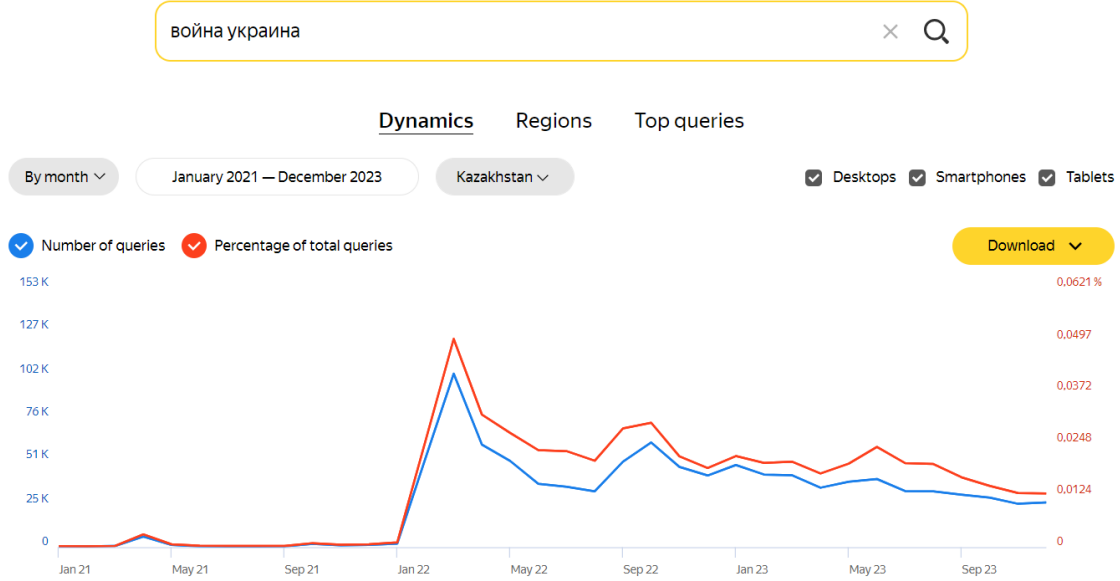
4 The Effect of Russian Language on Sanctions Circumvention

In this section, we study how Russian-language penetration affects the export of sanctioned goods from Kazakhstan to Russia and to other EAEU countries. Our aggregate export data includes 5610 observations at the region, month, product-class, and destination levels (17

provides only data on the relative popularity of search terms, making it difficult to assess actual search volumes or to make accurate comparisons over time or across keywords and regions. In contrast, Yandex Wordstat offers absolute search counts, providing more stable and reliable data for analysis.

²³Figure A7 of Appendix A presents the search results for the same phrase, “war in Ukraine”, in Kazakh. The number of searches does not exceed 130 per month, while queries in Russian reach tens of thousands, justifying our exclusive focus on search keywords in Russian. Moreover, Google Trends was unable to estimate the phrase’s popularity in Kazakh, displaying a message about insufficient data, as shown in Figure A8 of Appendix A.

Figure 4: Search for “War Ukraine” on Yandex



regions, 33 months, 5 product classes, 2 destinations). This section tests Prediction 4 of our conceptual framework. If identification with the Russian-speaking world increased following the full-scale invasion in 2022, it should surface in conformity behavior, and the most Russian-speaking regions should increase trade in sanctioned products that Russia demands.

Figure 2 shows that before the full-scale invasion, exports of subsequently sanctioned goods from Kazakhstan to Russia were small and relatively stable. While non-sanctioned goods were exported to Russia at the level of 8.4 million USD on average per month-region, the export of goods that were subsequently sanctioned stood at around 2.4 million USD per month-region. In the months after the start of the full-scale invasion, these exports increased multiple times, reaching a spike of 20 million USD per month-region in September 2022, while the export of non-sanctioned goods remained stable at pre-invasion levels.

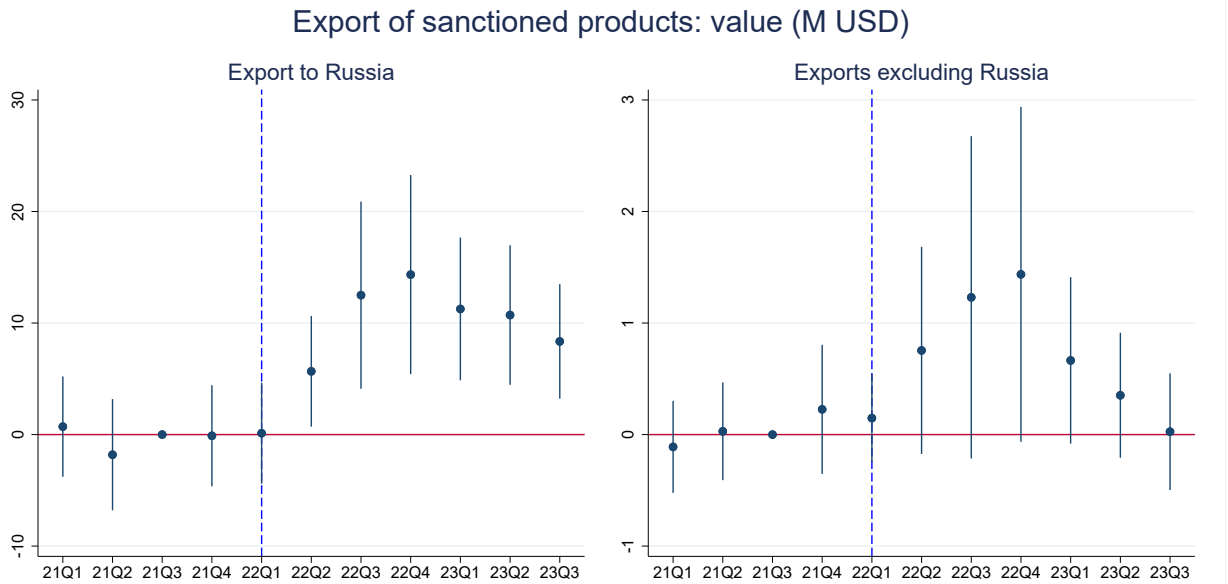
To begin with, we estimate the average effect of the full-scale invasion and the sanctions on exports of sanctioned goods to Russia. We use an event-study design, where the treatment group is composed of sanctioned products (electronics and machines), the control group includes all non-sanctioned products, and February 2022 is the treatment date. We consider

the following linear specification:

$$Y_{prt} = \sum_{\tau=21Q1}^{23Q3} \beta_{\tau} \cdot S_p \cdot \mathbf{I}(\tau = q_t) + \mu_p + \kappa_r + \lambda_t + \varepsilon_{prt} \quad (3)$$

The subscript r denotes region, t denotes time (year-month), and p product class. The outcome variable Y_{prt} is exports in million USD, along with the number of transactions and their log values. The binary variable S_p denotes sanctioned products, and q_t is the quarter of month t . Variables κ_r , λ_t , and μ_p are regional, time, and product class fixed effects, respectively. We consider exports to Russia and to other EAEU countries separately.

Figure 5: Value of export growth for sanctioned products: event-study



Note. The graph shows the event-study coefficients from (3). Blue dots show the coefficients β_{τ} . The 95% confidence intervals are calculated w.r.t. robust standard errors.

Figure 5 shows that after February 2022, exports of sanctioned goods to Russia increased by 9.8 million USD per region per month, while exports of these goods to other EAEU countries increased by only around 0.6 million USD (Table 2 Columns 1 and 2).²⁴ A similar pattern is observed for the number of transactions (Figure B1 of Appendix B) – the number of sanctioned product transactions to Russia increased by 468 per region-month, while it is only 42 for other

²⁴Throughout the paper, we call an estimate significant when it is significant at the 5 percent level.

EAEU countries.²⁵ Considering that the production output of electronics and machines has been very stable in Kazakhstan since 2021 (see Figure A5 in Appendix A for raw dynamics and Figure B4 of Appendix B for the event-study) and that a substantial spike in the export of these products to Russia occurred after February 2022 (Figure 2), this result suggests that Kazakh firms started to substantially re-export sanctioned goods to Russia after February 2022.

We now want to understand how the regional language gradient affects this re-export activity of sanctioned products – the focus of this section. We consider the following triple difference-in-differences (DID) specification:

$$\begin{aligned}
Y_{prt} = & \alpha \cdot S_p \cdot Treat_r + \beta \cdot S_p \cdot Post_t + \gamma \cdot S_p \cdot Post_t \cdot Treat_r + \delta \cdot Post_t \cdot Treat_r + \quad (4) \\
& \theta \cdot S_p \cdot Border_r + \phi \cdot S_p \cdot Post_t \cdot Border_r + \psi \cdot Post_t \cdot Border_r + \\
& \mu_p + \kappa_r + \lambda_t + \varepsilon_{prt}
\end{aligned}$$

Here, $Treat_r$ is the regional share of Russian speakers among ethnic Kazakhs from the 2021 Census (see the right panel of Figure 1) – the treatment exposure. The binary variables are $Post_t = 1$ for periods after February 2022, $Border_r = 1$ for regions with a common border with Russia, $S_p = 1$ for sanctioned products.

The coefficient α measures differences in the export of (not-yet) sanctioned products before February 2022 depending on the treatment exposure. The coefficient β measures the change in export of sanctioned goods after February 2022 for a counterfactual region with zero treatment exposure, while the coefficient γ captures the additional contribution of treatment exposure to this effect. The coefficient δ captures how treatment exposure changed exports of non-sanctioned products to Russia after February 2022. The second line of controls in (4) absorbs post-invasion changes in trade in both sanctioned and non-sanctioned products of border regions with Russia. In this specification, our focus is on $\beta + \gamma \cdot Treat_r$ – the marginal effect of (4) by $S_p \cdot Post_t$ for different levels of $Treat_r$ (net of border effect), which measures the change in exports of sanctioned products after February 2022 as the share of Russian-speaking

Table 2: The effect of Russian language on sanctions circumvention

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
	Dependent variable: export value (in M USD)					
Sanctioned products x Treatment exposure (α)			-7.12 (5.08)	0.042 (0.33)		
Sanctioned products x Post-invasion (β)	9.79*** (1.41)	0.63*** (0.21)	-3.23 (3.35)	0.41 (0.38)		
Sanctioned products x Post- invasion x Treatment exposure (γ)			39.4*** (9.04)	1.43*** (0.50)	26.7*** (6.74)	-0.23 (0.42)
Post-invasion x Treatment exposure (δ)			-3.00 (5.28)	-0.72** (0.32)		
Sanctioned products x Border region (θ)			-1.33 (1.51)	-0.080 (0.15)		
Sanctioned products x Post- invasion x Border region (ϕ)			-7.46*** (2.62)	-0.96*** (0.32)		
Post-invasion x Border region (ψ)			-2.98** (1.44)	-0.048 (0.12)		
Constant	19.5*** (0.95)	1.64*** (0.065)	20.8*** (1.66)	1.82*** (0.11)	1.99 (1.62)	1.18*** (0.14)
Observations	2,805	2,805	2,805	2,805	561	561
R-squared	0.301	0.435	0.311	0.440	0.524	0.747
Year-Month FE	Y	Y	Y	Y	Y	Y
Region FE	Y	Y	Y	Y	Y	Y
Product FE	Y	Y	Y	Y	N	N
Products	All	All	All	All	Sanctioned	Sanctioned
Sample	All regions	All regions	All regions	All regions	All regions	All regions
Export to	Russia	Non-Russia	Russia	Non-Russia	Russia	Non-Russia

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Columns 1 and 2 show the results of DID estimation of (3). Columns 3 and 4 show the results of a triple DID estimation of (4), with the regional share of Russian-speaking Kazakhs as the treatment exposure. Columns 5 and 6 show the results for (7). Odd columns show exports to Russia; even columns show exports to other EAEU countries. The dependent variable is the export value (in M USD) for a product class in a region-month. Robust standard errors are in parentheses.

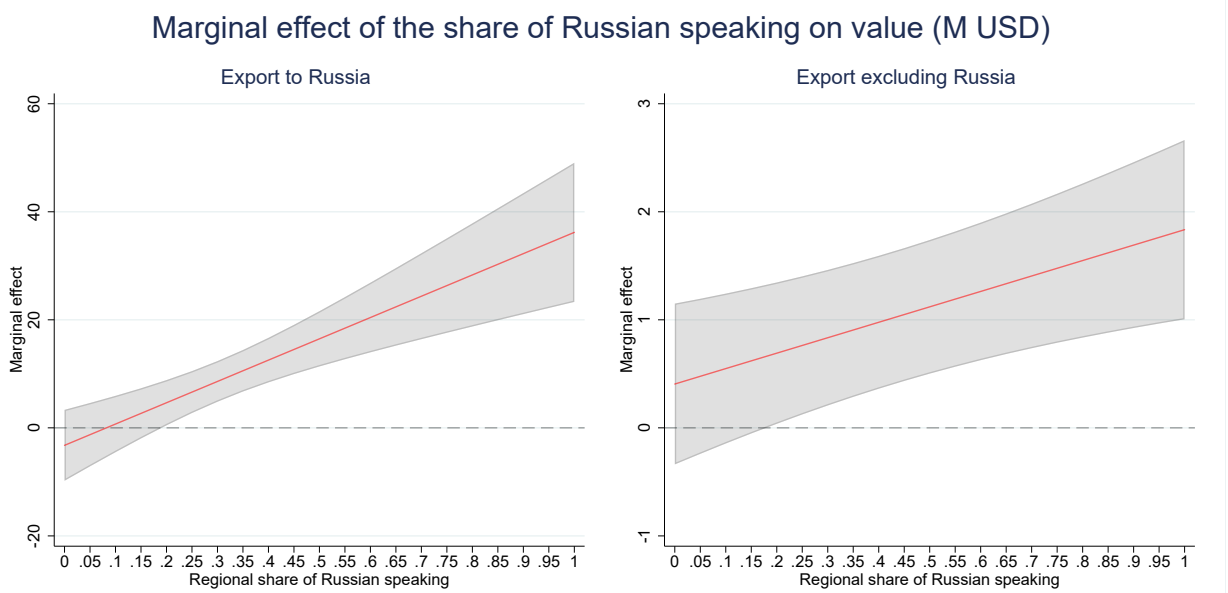
Kazakhs varies.

Table 2 presents the results for model (4) for the value of exports to Russia (Column 3) and to other EAEU countries (Column 4) as an outcome variable. The coefficient β is insignificant, suggesting that in a counterfactual region with a zero share of Russian-speaking Kazakhs, the full-scale invasion would have an insignificant effect on the export of sanctioned goods to Russia. The coefficient γ , on the other hand, is positive and significant for exports to Russia and to other EAEU countries, but the magnitude is substantially different. It suggests that exports of sanctioned products to Russia after February 2022 increase disproportionately

²⁵Figure B2 shows a similar pattern for the log of the export value and Figure B3 for the log of transactions.

from regions with a high share of Russian-speaking Kazakhs. This is consistent with the notion that supplying Russia is conformity to the interests of the group one identifies with (Prediction 4) rather than a uniform response to a profit opportunity. This becomes even more evident when we plot $\beta + \gamma \cdot Treat$ as a function of $Treat$ (Figure 6) – higher treatment exposure increases exports of sanctioned products to Russia far more than to EAEU countries after February 2022. Figure B5 of Appendix B shows the estimates of $\beta + \gamma \cdot Treat$ for the number of transactions as a dependent variable.²⁶ After February 2022, transaction growth increases with treatment exposure for exports to both Russia and other EAEU countries, but the effect is an order of magnitude larger for exports to Russia.

Figure 6: Value of export growth for sanctioned products by regional share of Russian-speaking Kazakhs



Note. The figure shows $\beta + \gamma \cdot Treat$ from the specification (4), where β and γ are taken from Columns 3 and 4 of Table 2. The dependent variable is the value of exports for a product class in a region-month (M USD). The regional share of Russian-speaking Kazakhs is based on the 2021 Census.

To see the dynamics of the treatment exposure and to test for parallel trends, we consider the following event-study analog of (4):

²⁶Figures B6 and B7 show estimates for the log of export value and the log of the transaction number.

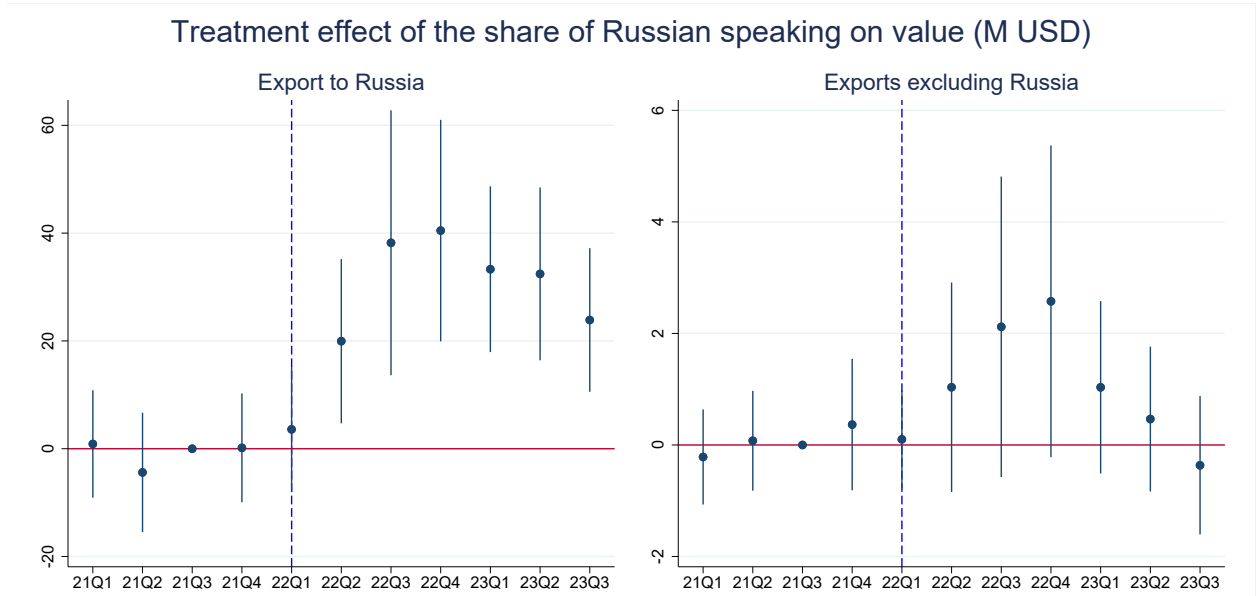
$$Y_{prt} = \alpha \cdot S_p \cdot Treat_r + \beta \cdot S_p \cdot Post_t + \sum_{\tau=21Q1}^{23Q3} \gamma_{\tau} S_p I(\tau = q_t) Treat_r + \delta \cdot Post_t \cdot Treat_r + \quad (5)$$

$$\theta \cdot S_p \cdot Border_r + \phi \cdot S_p \cdot Post_t \cdot Border_r + \psi \cdot Post_t \cdot Border_r +$$

$$\mu_p + \kappa_r + \lambda_t + \varepsilon_{prt}$$

Figure 7 shows that the parallel trends assumption holds for the export value as a dependent variable. Moreover, the treatment exposure had the strongest effect on the re-export of sanctioned products to Russia in the second half of 2022. Figure B8 shows a similar pattern for the number of transactions as the outcome variable.

Figure 7: Event study for the value of export growth by regional share of Russian-speaking Kazakhs



Note. The figure shows coefficients γ_{τ} from (5). The dependent variable is the export value (M USD) for a product class in a region-month.

Next, we also estimate the extension of model (5) to separate the effect of Russian language

from Russian ethnicity:

$$\begin{aligned}
Y_{prt} = & \alpha \cdot S_p \cdot Treat_r + \beta \cdot S_p \cdot Post_t + \sum_{\tau=21Q1}^{23Q3} \gamma_{\tau} S_p I(\tau = q_t) Treat_r + \delta \cdot Post_t \cdot Treat_r + \quad (6) \\
& \theta \cdot S_p \cdot Border_r + \phi \cdot S_p \cdot Post_t \cdot Border_r + \psi \cdot Post_t \cdot Border_r + \\
& \eta \cdot S_p \cdot RusEth_r + \nu \cdot S_p \cdot Post_t \cdot RusEth_r + \rho \cdot Post_t \cdot RusEth_r + \\
& \mu_p + \kappa_r + \lambda_t + \varepsilon_{prt}
\end{aligned}$$

Here $RusEth_r$ is the share of ethnic Russians in region r . All other variables are similar to equation (4). Figures B9 and B10 show the results for the export value and its log as an outcome variable; Figures B11 and B12 show the results for the number of transactions and its log. The results are similar to those of the baseline model (4). This suggests that Russian ethnicity does not entirely explain our main results.

Most circumvention of export sanctions runs through friendly third countries, which offset roughly two-thirds of the drop in Russia’s imports (Egorov et al. 2026; Chupilkin et al. 2026). In this literature, circumvention is arbitrage. Sanctions make goods scarce in Russia, and firms in neutral neighbors profit from supplying them. Within Kazakhstan, all regions face the same Russian demand, and (4) absorbs border logistics through $Border_r$ and its interaction with time and product type. Profit alone therefore predicts uptake everywhere, with $\beta > 0$ and $\gamma \approx 0$. We find a steep language gradient instead. Within the observed range of the treatment exposure (0.27 to 0.66), predicted export growth of sanctioned products rises from about 7.6 to about 22.9 million USD per region-month (Figure 6). The insignificant α (Table 2, Columns 3 and 4) indicates that Russian-speaking regions had no head start in this trade before the war. As Figures B15 and B16 show, the same results hold if we use the regional share of Russian speakers among ethnic Kazakh firm managers from the 2020-2021 CAB waves as the treatment exposure, suggesting that the language as a social phenomenon also frames the businessmen’s views with respect to war and sanctions.²⁷

Finally, we restrict our attention to the subsample of export data, including only sanctioned

²⁷The correlation between the regional share of Russian speakers among Kazakhs from Census 2021 and the regional share of Russian speakers among Kazakh firm managers from 2020-2021 CAB waves is 0.76.

goods, and estimate a DID specification similar to the one in [Korovkin and Makarin \(2023\)](#):

$$Y_{rt} = \gamma \cdot Post_t \cdot Treat_r + \kappa_r + \lambda_t + \varepsilon_{rt}, \quad (7)$$

with the notation similar to the one in equation (4).²⁸ In this specification, coefficient γ identifies the differential export growth of sanctioned goods after February 2022 between regions with high and low degrees of treatment exposure. Columns 5 and 6 of Table 2 show the results for the value of export to Russia and to other EAEU countries.²⁹ Similar to our previous findings, growth in exports of sanctioned goods to Russia increased from regions with a higher share of Russian-speaking Kazakhs, while it did not for the export to other EAEU countries. An increase in the regional share of Russian-speaking Kazakhs from the minimum (0.27) to the maximum (0.66) increases additional exports by 10.3 million USD per month and region. In line with our previous findings, these results suggest that Russian-speaking Kazakhs significantly increased the re-export of sanctioned products to Russia after the full-scale invasion of Ukraine. Taken together, the trade evidence supports Prediction 4.

5 Mechanisms

We showed that Russian-language penetration increases sanctions circumvention. This section establishes the mechanism. We proceed in three steps that mirror the predictions of Section 2. A pre-registered experiment shows that a language cue alone activates identification with Russia and improves attitudes toward it. The CAB waves show that after the invasion, language polarized people’s attitudes toward Russia. Yandex search data show how media consumption follows the shift and stabilizes it. A final subsection discusses the alternative explanations.

5.1 The Effect of Russian Language on Attitudes to Sanctions

Our difference-in-differences estimates show that Russian-speaking regions re-exported more sanctioned goods to Russia after February 2022. We identify this effect at the regional level,

²⁸One can think of regions with low and high treatment exposure as control and treatment groups, respectively, in a binary treatment setting.

²⁹Figures B13 and B14 show the result of the event-study analysis for (7) for the export value and the number of transactions.

using cross-regional variation in the share of Russian speakers. Even with specifications that control for border regions and Russian ethnicity penetration interacted with the post-invasion period, and despite parallel trends, regional Russian-language penetration is not randomly distributed. In this section, we provide complementary evidence at the individual level, where language itself is randomly assigned, reinforcing that the effect of Russian-speaking identity is causal.

Let binary Z_i indicate that respondent i was randomly addressed in Russian on the introductory page, and let R_i indicate that i *completed* the survey in Russian. For an attitude Y_i , our structural equation of interest is

$$Y_i = \beta_0 + \beta_1 R_i + \mathbf{X}_i \delta + \varepsilon_i, \quad (8)$$

where β_1 measures the causal effect of completing the survey in Russian on the attitude, and $\mathbf{X}_i$ is a vector of controls including: ethnicity, gender, age, number of children, education, occupation, household welfare, internet-use intensity, a rural indicator, and regional fixed effects. Because R_i is a choice and hence potentially endogenous, we instrument it with the randomly assigned language of the introductory page Z_i . Therefore, the first stage is

$$R_i = \pi_0 + \pi_1 Z_i + \mathbf{X}_i \gamma + \nu_i, \quad (9)$$

so that π_1 captures how strongly the assigned language shifts the language respondents actually use. Since Z_i is randomized, we also report the reduced form estimate,

$$Y_i = \alpha_0 + \alpha_1 Z_i + \mathbf{X}_i \theta + u_i, \quad (10)$$

where α_1 is the effect of being primed with a Russian-speaking identity. We report each specification both with and without controls $\mathbf{X}_i$. The specification with controls absorbs residual variation and improves precision. Standard errors are heteroskedasticity-robust.

We begin with the effect of the randomized introductory-page language on the language respondents actually use. Table 3 shows the results. Being addressed in Russian raises the probability of *completing* the survey in Russian by about 22 percentage points (Columns 1 and

2). The randomized introductory page also increases stated use of Russian with friends and colleagues and for news (Columns 5 to 8), but not at home (Columns 3 and 4), where language is most habitual and hardest to move. Completing the survey in Russian is a behavior rather than an answer to a question, so this is our central measure of a Russian-speaking identity. Moreover, our randomized instrument is a strong predictor of the survey language,³⁰ keeping us safely beyond the weak instrument problem (Andrews et al. 2019). The randomly assigned introductory language is also a valid instrument for the survey language by construction of randomization (for balance statistics see Table C2), so IV estimates the compliers' average causal effect of survey language on attitudes.

Table 3: Russian introductory page and Russian language use

VARIABLES	(1) Survey in Russian	(2) Survey in Russian	(3) Russian at home	(4) Russian at home	(5) Russian with friends	(6) Russian with friends	(7) News in Russian	(8) News in Russian
Intro page in Russian	0.23*** (0.014)	0.22*** (0.011)	0.012 (0.014)	0.0022 (0.0091)	0.031** (0.014)	0.023** (0.0097)	0.032** (0.014)	0.023** (0.010)
Constant	0.42*** (0.0098)	0.23*** (0.042)	0.37*** (0.0096)	0.16*** (0.035)	0.40*** (0.0097)	0.079** (0.038)	0.43*** (0.0098)	0.076* (0.041)
Observations	5,025	5,025	5,025	5,025	5,025	5,025	5,025	5,025
R-squared	0.051	0.443	0.000	0.559	0.001	0.519	0.001	0.470
Demographics	N	Y	N	Y	N	Y	N	Y
Region FE	N	Y	N	Y	N	Y	N	Y
Estimat.	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The table shows the effect of randomized introductory-page language on Russian-language use across different circumstances. In Columns 1-2, the dependent binary variable equals 1 if the survey was completed in Russian. In Columns 3-4, the dependent binary variable equals 1 if the respondent's primary language for communicating at home is Russian. In Columns 5-6, the dependent binary variable equals 1 if the respondent's primary language for communicating with friends and colleagues is Russian. In Columns 7-8, the dependent binary variable equals 1 if the respondent's primary language for news consumption is Russian. The wording of questions is in Section 3.2. Odd columns do not have controls. Even columns include additional controls: ethnicity, gender, age, number of children, education, occupation, household welfare, internet use intensity, a rural binary indicator, and regional fixed effects. Robust standard errors are in parentheses.

Table 4 reports reduced-form regressions of attitudes toward Russian affairs on the randomly assigned introductory language. Respondents who receive the Russian introductory page are 3 p.p. (12 percent) more likely to consider helping Russia evade sanctions justified

³⁰Kleibergen-Paap Wald rk F-statistic is above 260 in models without controls and above 390 in models with controls (see Table 5).

(Columns 1 and 2), 4 p.p. (17 percent) more likely to justify the invasion of Ukraine (Columns 3 and 4), and hold a 2.7% more favorable opinion of Russia (Columns 5 and 6). Table 5 reports the resulting IV estimates. Completing the survey in Russian increases justification for helping Russia evade sanctions by 15 p.p. (68%), and for the invasion by 18 p.p. (more than doubling), and raises overall opinion of Russia by 12%.

Table 4: Reduced form: Russian introductory page and attitudes toward Russian affairs

VARIABLES	(1) Sanctions evasion	(2) Sanctions evasion	(3) Invasion/SMO	(4) Invasion/SMO	(5) Opinion of Russia	(6) Opinion of Russia
Intro page in Russian	0.034*** (0.013)	0.030** (0.013)	0.041*** (0.012)	0.039*** (0.012)	0.091*** (0.031)	0.080*** (0.029)
Constant	0.28*** (0.0089)	0.47*** (0.051)	0.24*** (0.0084)	0.45*** (0.047)	3.32*** (0.022)	3.78*** (0.11)
Observations	5,025	5,025	5,025	5,025	4,334	4,334
R-squared	0.001	0.072	0.002	0.105	0.002	0.137
Demographics	N	Y	N	Y	N	Y
Region FE	N	Y	N	Y	N	Y
Estimat.	OLS	OLS	OLS	OLS	OLS	OLS

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The table shows the effect of randomized introductory-page language on attitudes toward Russian affairs. In Columns 1-2, the dependent binary variable equals 1 if helping Russia in sanctions evasion is “justified”. In Columns 3-4, the dependent binary variable equals 1 if the invasion of Ukraine/Special Military Operation is “justified”. In Columns 5-6, the dependent scale variable is the opinion of Russia (from 1 – “Very negative” to 5 – “Very positive”). The wording of questions is in Section 3.2. Odd columns do not have controls. Even columns include additional controls: ethnicity, gender, age, number of children, education, occupation, household welfare, internet use intensity, a rural binary indicator, and regional fixed effects. Robust standard errors are in parentheses. Table C3 shows results for attitudes toward non-Russian affairs.

According to Prediction 3, the cue activates a Russian-speaking identity, which implies that randomly assigned introductory pages should activate Russia-related attitudes but not attitudes toward other affairs. Table C3 (reduced form) and Table C4 (IV) in Appendix C show that neither the introductory page nor the survey in Russian has a significant effect on opinions of China or the United States, trust in national government, the justifiability of participating in protests, or overworking.

Table 5: The effect of Russian survey language on attitudes toward Russian affairs

VARIABLES	(1) Sanctions evasion	(2) Sanctions evasion	(3) Invasion/SMO	(4) Invasion/SMO	(5) Opinion of Russia	(6) Opinion of Russia
Survey in Russian	0.15** (0.059)	0.14** (0.059)	0.18*** (0.057)	0.18*** (0.055)	0.38*** (0.13)	0.34*** (0.12)
Constant	0.22*** (0.032)	0.44*** (0.054)	0.16*** (0.031)	0.41*** (0.052)	3.17*** (0.068)	3.70*** (0.12)
Observations	5,025	5,025	5,025	5,025	4,334	4,334
Demographics	N	Y	N	Y	N	Y
Region FE	N	Y	N	Y	N	Y
Estimat.	IV	IV	IV	IV	IV	IV
Kleibergen-Paap Wald rk F	269.3	408.7	269.3	408.7	266.4	391.5

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The table shows the IV results of the effect of completing the survey in Russian on attitudes toward Russian affairs. The dependent variables (and control variables in the even columns) are the same as in Table 4. The heteroskedasticity-robust F-statistic shows that the instrument is strong (≥ 20) for standard inference (Andrews et al. 2019). Robust standard errors are in parentheses. Table C4 shows results for attitudes toward non-Russian affairs. Table C5 shows results for Russian language use across different circumstances.

Finally, a randomized introductory page in Russian may affect not only how people think of Russia but also their perception of how prevalent Russian is in their daily lives. We have already shown the reduced-form results in Table 3 for Russian language use in different circumstances. Table C5 in Appendix C shows IV results, showing that completing the survey in Russian raises respondents' reported use of Russian with friends and for news. A short, random exposure to Russian thus shifts both behavior and stated language use, consistent with the view that the cue activates an identity. That the effect appears for language used with friends and for news, but not at home, fits this reading. The cue activates the more public, identity-expressive domains of language use, while the most habitual domain is slower to move. Taken together, these results support Prediction 3 and establish the causal relation. The effect is specific to Russian affairs, expressed in revealed behavior as well as in stated preferences. It operates even in the absence of any media exposure, identifying a standalone role for language. These attitudes, in turn, induce the re-export of sanctioned goods documented in Section 4.

5.2 The Polarizing Effect of the Russian Language

So far, we have shown that Russian-speaking identity increases pro-Kremlin views in Fall 2023. The relevant question is whether Russian speakers have always been more pro-Russian than non-Russian speakers, as they could share long-term historical and cultural ties with Russia, or whether the full-scale invasion triggered an identity affiliation. The CAB data allow us to test Predictions 1 and 2 together. The war should lower the average opinion of Russia for everyone, because the status shock is common to everyone (decrease of S_R). At the same time, it should open a gap by language, because the salience shock magnifies differences that already exist. To examine this argument, we use questions on attitudes to Russia, the US, and China from eight CAB waves from 2020 to 2023, allowing us to estimate the following DID specification:

$$y_{irt} = \alpha Treat_r + \beta Post_t + \gamma Treat_r \cdot Post_t + \mathbf{X}_{irt}\delta + \epsilon_{irt} \quad (11)$$

Here, i denotes a respondent from region r and survey wave t . The outcome variable y_{irt} is the opinion of Russia/US/China on a 4-level Likert scale (1 – “very unfavorable”, 4 – “very favorable”), $Treat_r$ is the regional share of Russian-speaking Kazakhs from the 2021 Census, and $Post_t$ is the binary indicator for waves after February 2022. The vector $\mathbf{X}_{irt}$ includes the following set of control variables: gender, five age categories, a binary indicator for higher education, five categories of household size, a binary indicator for urban settlement, Russian and Kazakh binary indicators of ethnicity, as well as the interviewer’s assessment of the respondent’s survey comprehension and comfort with answering the survey. The control for individual ethnicity enables us to separate ethnic identity from the regional-level language effect.

The coefficient α captures the language gap in y_{irt} before February 2022, depending on the regional share of Russian-speaking Kazakhs. Coefficient β shows the trend in y_{irt} after February 2022 for a hypothetical region with zero Russian-speaking Kazakhs. Under the parallel trends assumption, the coefficient γ measures divergence from the trend, depending on the share of Russian-speaking Kazakhs after February 2022, i.e., the effect of the Russian-language penetration on y_{irt} after February 2022.

We also consider the extension of (11) by introducing the region (κ_r) and survey wave (λ_t) fixed effects, to take into account all the variation (between regions and over time) that is not related to the Russian-language penetration and the full-scale invasion.

$$y_{irt} = \gamma Treat_r \cdot Post_t + \mathbf{X}_{irt}\delta + \kappa_r + \lambda_t + \epsilon_{irt} \quad (12)$$

Table 6: Opinion of Russia, the US, and China

VARIABLES	(1) Opinion of Russia	(2) Opinion of Russia	(3) Opinion of the US	(4) Opinion of the US	(5) Opinion of China	(6) Opinion of China
Treatment exposure (α)	-0.12* (0.069)		0.052 (0.087)		0.48*** (0.084)	
Post-invasion (β)	-0.57*** (0.047)		-0.020 (0.054)		0.14*** (0.051)	
Treatment exposure x Post-invasion (γ)	0.44*** (0.11)	0.47*** (0.11)	-0.073 (0.13)	-0.079 (0.13)	0.20* (0.12)	0.22* (0.12)
Constant	3.22*** (0.040)	3.12*** (0.099)	2.67*** (0.050)	2.68*** (0.11)	2.46*** (0.047)	2.76*** (0.10)
Observations	11,934	11,934	10,759	10,759	11,497	11,497
R-squared	0.122	0.137	0.111	0.117	0.120	0.131
Time FE	N	Y	N	Y	N	Y
Region FE	N	Y	N	Y	N	Y
Mean DV	2.808	2.808	2.625	2.625	2.498	2.498

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The odd columns show results of regression (11), even columns – (12) for scaled dependent variables: “Opinion of Russia” (Columns 1-2), “Opinion of the US” (Columns 3-4), “Opinion of China” (Columns 5-6). The treatment exposure is the regional share of Russian-speaking Kazakhs from the 2021 Census. The set of control variables includes gender, five age categories, a binary indicator for higher education, five categories of household size, a binary indicator for urban settlement, Russian and Kazakh binary indicators of ethnicity, as well as the interviewer’s assessment of the respondent’s survey comprehension and comfort with answering the survey. Robust standard errors are in parentheses.

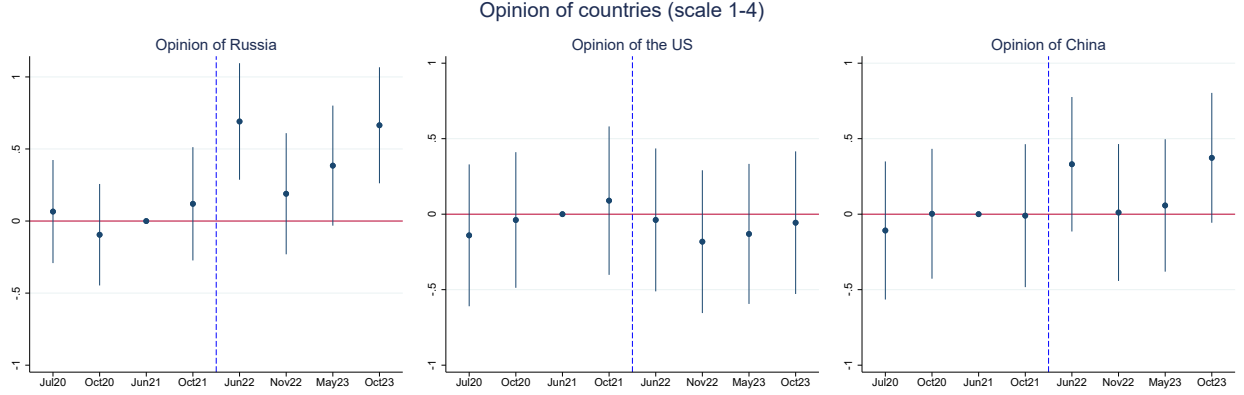
Table 6 shows the estimates for (11) (Columns 1, 3, and 5) and (12) (Columns 2, 4, and 6). Coefficient α is insignificant for the opinions of Russia and the US, and positive for China. This suggests that before the full-scale invasion, respondents from regions with a higher share of Russian-speaking Kazakhs did not have more positive views of Russia and the US, but did

have more positive views of China. Coefficient β is negative for Russia, insignificant for the US, and positive for China, suggesting that the view of Russia (China) substantially deteriorated (improved) in regions with a low share of Russian-speaking Kazakhs. This negative and Russia-specific β corresponds to Prediction 1. The war lowers Russia’s status for all residents, while opinions of the US do not fall. Finally, Russian-language penetration positively affected only the opinion of Russia after February 2022 (γ), with the opposite sign to β , suggesting polarization of opinions about Russia after February 2022 between regions with high and low shares of Russian-speaking Kazakhs. The positive γ works against this common decline and corresponds to the widening gap of Prediction 2. The insignificant α confirms the second part of Prediction 2. Before the war, language carried little attention weight and predicted opinions of Russia only modestly. Because the treatment is the share of Russian-speaking *Kazakhs* and we control for individual ethnicity, the gap opens along language rather than ethnicity, as Prediction 2 suggests.

Figure 8 shows event-study coefficients for the DID specification (12). For all three outcome variables, there are parallel trends before February 2022, and only “Opinion of Russia” is positively affected by the regional share of Russian-speaking Kazakhs after February 2022. The gap also does not fade over the sample period, as the event-study coefficients remain elevated through late 2023. Table D1 and Figure D1 show the DID and event-study estimates for the two language outcome variables: “Russian at home” and “Survey in Russian”. The 2021 Census regional share of Russian-speaking Kazakhs does not affect the probability of using the Russian language at home after February 2022, while it increases the probability of completing surveys in Russian after February 2022, reinforcing the polarization argument rather than consolidation, as was shown for Ukraine (Abramenko et al. 2026).

Since the language used at home at the individual level is unaffected by the regional share of Russian-speaking Kazakhs after February 2022, we use it as the treatment exposure in the

Figure 8: Event-study for Opinions of Russia, the US, and China by the regional share of Russian-speaking Kazakhs



Note. The figure shows event-study coefficients γ_t for the DID specification (12) and dependent variables: “Opinion of Russia” (left), “Opinion of the US” (middle), “Opinion of China” (right).

extended specification of (12):³¹

$$y_{irt} = \alpha Treat_{it} + \gamma Treat_{it} \cdot Post_t + \mathbf{X}_{irt} \delta + \kappa_r + \lambda_t + \epsilon_{irt}. \quad (13)$$

Here, $Treat_{it}$ is 1 if the respondent speaks Russian at home as the main language and 0 otherwise. Table D2 shows the results of the estimation for (13) for all respondents and firm managers, while Figures D2 and D3 show the event-study analysis. The results are very close to those in Table 6, showing polarization in attitude to Russia by language spoken at home among the general population and firm managers. For waves after 2022, Appendix D.1 also shows that Russian-speaking respondents are more likely to justify the invasion and blame Western countries as most responsible for it. This association holds for the general population and firm managers, and it reinforces our causal experimental result shown in Section 5.1, though the surveys are conducted by different companies and in different survey modes.

Overall, the CAB evidence supports Predictions 1 and 2. The war lowered everyone’s opinion of Russia and at the same time polarized regions along language. The argument that Russian speakers have simply always been closer to the Russian world is therefore not

³¹This specification does not have a causal interpretation per se, as the language used at home is wave- and individual-specific. Nevertheless, combined with results from Table D1 and Figure D1, it can have a quasi-causal interpretation if the event-study shows no dynamics for the gap in attitudes before the invasion by individual language spoken at home.

supported. In the next section, we turn to the media demand channel of Prediction 5. Those who newly identify with R should demand more Russian-language media, which in turn feeds the polarization back.

5.3 The Effect of Russian Language on Media Choice

So far, we have shown that the Russian language itself increases justification of the invasion, sanctions circumvention, and opinion of Russia. We also demonstrated that this gap in pro-Russian opinion emerged only after February 2022. This section tests the first component of Prediction 5. Demand for Russian-language media should rise after February 2022 as a further expression of conformity by those who newly identify with the Russian-speaking world.

To test this hypothesis, we use Yandex search statistics at the month-by-region level in this section. To see whether Russian-language penetration affected the frequency of searches for war-related information, we use the 2021 Census share of Russian-speaking Kazakhs at the regional level and match it to regional search statistics from Yandex. We consider the following event-study specification:

$$y_{rt} = \sum_{\tau} \gamma_{\tau} \mathbf{I}(\tau = t) Treat_r + \kappa_r + \lambda_t + \epsilon_{rt} \quad (14)$$

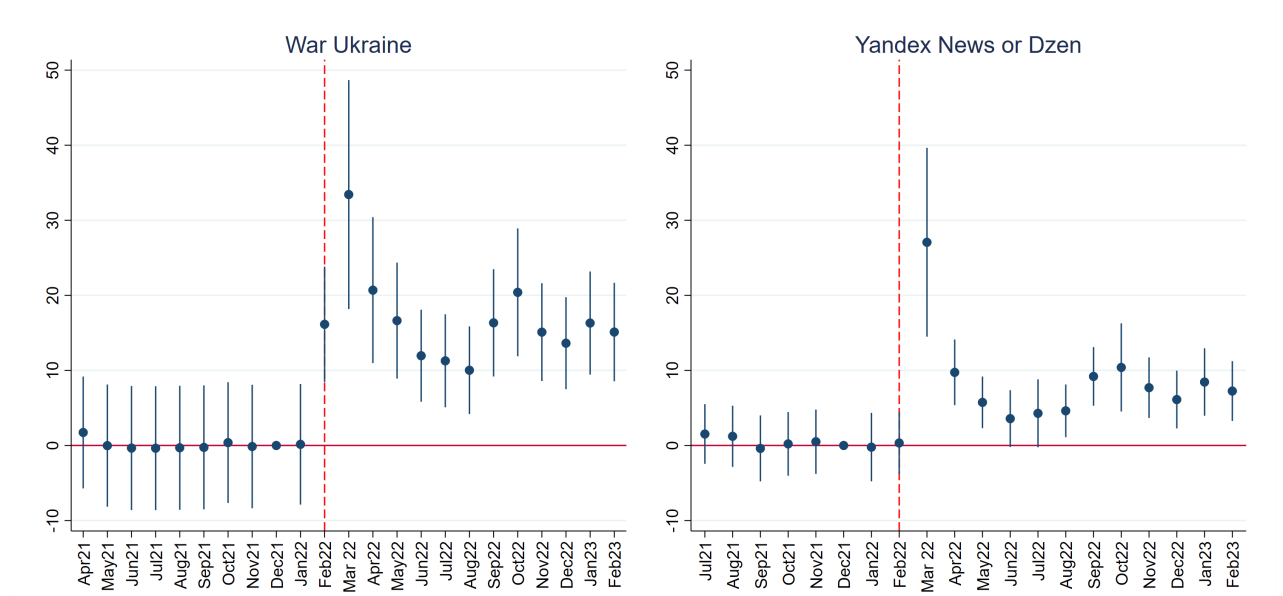
The dependent variable $y_{rt} = \frac{\text{Num. of searches in region } r \text{ in month } t}{\text{Population in region } r} \cdot 1000$ is the number of searches for a specific keyword per month-region per 1,000 inhabitants. $Treat_r$ is the share of Russian-speaking Kazakhs in region r according to the 2021 Census (Figure 1), κ_r is the region fixed effect, and λ_t is the year-month fixed effect. The coefficient γ_{τ} captures the differential effect of regional Russian-language penetration on the frequency of online searches in month t .

The left graph of Figure 9 shows the results for the keywords “vojna Ukraina” (“war Ukraine”, see also Figure 4).³² It shows that regions with a higher share of Russian speakers experienced a significant increase in searches for war-related keywords compared to regions with a lower share of Russian speakers, following the February 2022 event. Specifically, increasing the share of Russian speakers from 27% to 66% in a region (that is, from the lowest to the highest share in Kazakhstan) increases the number of searches for war-related keywords

³²The Yandex search statistics are insensitive to letter case, prefixes, suffixes, pronouns, and prepositions, as a search phrase is first lemmatized and stopwords are removed.

by 13.3 per 1,000 people.³³ This language gap spike in search for war-related information in Russian does not mean that people automatically ended up on pro-Kremlin websites. To investigate this further, we examine the search patterns for specific news aggregators and outlets.

Figure 9: Event-study coefficients for Yandex search “war Ukraine”, “Yandex news” or “Yandex Dzen”



Note. The figure shows coefficients γ_τ from (14) over time. The treatment exposure is the regional share of Russian-speaking Kazakhs. The outcome variable is the number of keyword searches of a keyword in Yandex per 1,000 regional inhabitants.

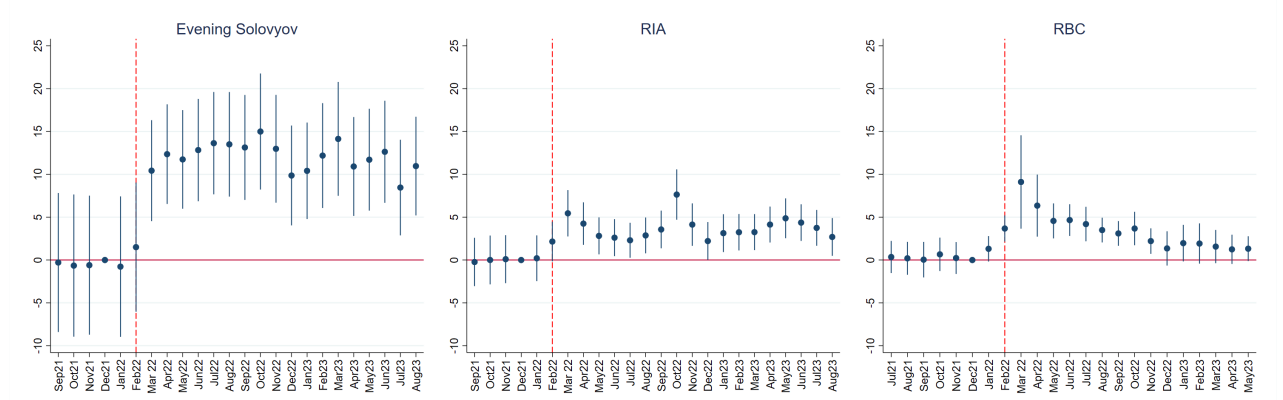
First, we estimate equation (14) for searches for Yandex Dzen or Yandex News – the largest pro-Kremlin news aggregator.³⁴ The right graph of Figure 9 shows that the effect of the regional share of Russian speakers on searches for Yandex News is comparable to that for “war Ukraine”. This suggests that Yandex News is an important source of news in the Russian-speaking regions of Kazakhstan after February 2022. Figure 10 shows that other pro-Kremlin TV shows and media outlets, such as “Evening with Solovyov”, RIA News, and

³³Figure E2 in Appendix E shows that the large-scale protests that occurred in Kazakhstan in January 2022 – the most significant protests that ever happened in Kazakhstan’s history as an independent country – led to a much smaller language gap in the search for information, highlighting that the interest of the Russian-speaking population in searching for information induced by Russia’s full-scale invasion of Ukraine was truly exceptional.

³⁴Yandex News was acquired by the Russian state-owned social media VK in 2022, and renamed Yandex Dzen.

RBC News, also experienced a significant surge in Russian-speaking regions after February 2022.

Figure 10: Event-study coefficients for Yandex search of pro-Kremlin news outlets



Note. The figure shows coefficients γ_τ from (14) over time. The treatment exposure is the regional share of Russian-speaking Kazakhs. The outcome variable is the number of keyword searches of a keyword in Yandex per 1,000 regional inhabitants.

Searches for Russian-language versions of the BBC and Euronews (the most popular pro-Western outlets in Kazakhstan), on the other hand, showed only a very small spike in Russian-speaking regions, suggesting that Western outlets were unable to compete in Kazakhstan’s Russian-language information space (Figure E1). Finally, Figure E3 shows that the effect of Russian-language penetration on news searches after February 2022 is specific to the war in Ukraine, and that other generic searches, such as weather, online games, or movies, are not connected to this event. All in all, the search data confirm the media demand component of Prediction 5. Russian-language penetration sharply increases searches for pro-Kremlin news after February 2022, and specifically for war-related content.

5.4 Discussion

Each of our data sources addresses a distinct threat to a causal reading of the others. The trade data show that Russian-language penetration increases the re-export of sanctioned products to Russia; the survey experiment highlights the exclusive effect of Russian-speaking identity; the sequence of the survey data shows the language-based polarization appears only after February 2022, and the online search data highlight media as one of the mechanisms. Together they support a simple reading in three steps. First, the war sorted the population into groups

along language, much as partisanship in the United States, Brexit, or the pandemic sorted other societies along a single line (Druckman et al. 2013; Hobolt and Tilley 2026; Gollwitzer et al. 2020), with the difference that the shock activated an old cleavage instead of creating a new one. Second, the groups processed the same war differently. People accept information that fits their group with little scrutiny and argue against information that does not (Kunda 1990; Taber and Lodge 2006; Enikolopov et al. 2026), so attitudes toward Russia and toward sanctions evasion, and with them the re-export of sanctioned goods, shifted downstream of identity, consistent with citizens adjusting attitudes to identities rather than the reverse (Pop-Eleches and Robertson 2018). Third, people sought out media congruent with the activated identity, so that media consumption is a consequence of the identity shift and not its cause. Each alternative reading below questions one of these steps, and the rest of this subsection asks how far it carries and what it leaves for identity to explain.

Profit offers the first alternative reading, since sanctions made goods scarce in Russia and the rerouting literature treats circumvention as arbitrage, so firms from anywhere in the customs union should collect the rents (Egorov et al. 2026). Our aggregate estimates are consistent with this rent-driven explanation. What arbitrage alone does not predict is the gradient, since predicted re-export growth is about three times larger in the most than in the least Russian-speaking region (Section 4). Language plausibly does lower the cost of finding and contracting with Russian buyers (Melitz 2008; Melitz and Toubal 2014). This variant predicts the same trade gradient as identity, so the trade data alone cannot reject it. The attitude evidence from Section 5.1 shows what transaction costs alone leave unexplained. A contracting advantage implies no reaction to a randomized language cue, yet the cue raises stated justification of sanctions evasion by 3 percentage points. It also implies no change in attitudes over time, yet a gap in opinions of Russia opens after February 2022 and persists. A last variant runs the causality backward, with the re-export boom producing the attitudes that justify it. In our experiment, however, the randomized cue moves attitudes with no transaction involved. Additionally, the attitude gap opens in the general population, few of whom trade, in the first CAB wave after the invasion and before the surge peaks in late 2022. We conclude that opinions moved first and trade followed, matching the broader pattern that citizens adjust their attitudes to their identities rather than the reverse (Pop-Eleches

and Robertson 2018). Profit explains why circumvention happens but identity predicts who engages in it.

A second concern is that the survey responses are cheap talk, because survey answers always carry an element of self-presentation and part of the cue effect may well be conformity. Conformity is not an alternative to our account, since in the framework identifying with a group means moving toward what the group expects (Shayo 2020; Jia and Persson 2021). The concern is rather that respondents say what a Russian-speaking audience wants to hear without holding the view. Three observations speak against this. First, the cue does not make respondents more agreeable in general, since opinions of China and the United States, trust in the national government, protest participation, and the work-hours placebo do not move. Second, there is no audience to please, because the survey was self-administered online with no interviewer present, and after a single introductory page respondents chose the survey language themselves. Demand effects in survey experiments of this kind are small (Mummolo and Peterson 2017). Third, the same pattern appears in a different survey mode. The Central Asia Barometer is run by a different company, and interviewers ask the questions by phone. There, Russian speakers are 11 percentage points more likely to justify the invasion, and Russian-speaking firm managers 9 percentage points (Appendix D.1). Firm managers are the group that decides on trade, so the attitudes we measure are held by those who could act on them.

A final alternative reverses the third step and attributes the attitude shift to persuasion by Russian media. This is the standard reading in work on Russian media abroad, where exposure to Russian television moves votes and attitudes (Peisakhin and Rozenas 2018; Enikolopov et al. 2011). Our framework implies the opposite order (Section 2). The war sorted the population along language through the identity mechanism, and once identity has shifted, media consumption is predictable from it. People who come to identify with the Russian-speaking world seek out its media. Media sit downstream of identity and can entrench the divide, but they do not create it. The link between identity and media is *confirmation bias*, the tendency to seek out and interpret information in a way that confirms what one already holds (Nickerson 1998; Epley and Gilovich 2016). Before the invasion there was little to confirm, since language groups did not differ in their views of Russia. Social identity creates

the condition for the bias. Once people identify with a group, they have a stake in the group being right (Tajfel and Turner 1986), and the bias then works in its two familiar forms. The first is biased search. Section 5.3 shows that Russian-speaking regions turned to pro-Kremlin outlets after the invasion, while generic searches did not move. The second is biased interpretation. Exposure to a foreign power’s outlet persuades those already close to it and repels the rest, deepening the divide it meets rather than creating a new one (Kao 2026). When outlets of opposite slant both deepen the divide they meet, the direction of the effect comes from the receiver’s identity and not from the content (Bisgaard and Slothuus 2018; Slothuus and Bisgaard 2021). Inside a like-minded online network, news that contradicts the group does not change minds, because the network supplies the counterarguments (Enikolopov et al. 2026). The experiment is consistent with this order, since the randomized cue moves attitudes with no media exposure. Media therefore explain why the divide persists through late 2023, and identity explains why it opened in the first place.

Beyond Kazakhstan, our results imply that a state that stays neutral in a war cannot count on its citizens to stay neutral as well. Kazakhstan’s government kept its distance from the war, yet trade aggregates the sides its citizens take, so a bystander in the diplomatic sense can be a participant in the economic one. The narrative under which a war is fought can therefore have material returns, because the claim to protect Russian speakers abroad may have contributed to activating the very identity that then helped supply the sanctioned goods. The effectiveness of export controls thus depends on the identity geography of third countries, a margin outside the prices and enforcement on which the rerouting literature focuses (Egorov et al. 2026). This geography can lie dormant. Before the invasion, language predicted neither attitudes toward Russia nor trade in the goods later sanctioned, so cohesion measured in peacetime says little about where a society divides under an external shock. This is what unfinished nation-building means in our setting. Rulers homogenize through language and schooling because a common identity raises the willingness to fight, and homogenization matters most for the citizens closest, in preference or language, to the rival capital (Alesina et al. 2020). Three decades of Kazakh-medium schooling and a planned Latin script, alongside a constitution that keeps Russian on an equal footing in government, have not closed that distance for a large share of the population. It carried no weight until a war fought in the name of the language gave it one.

Nation-building is usually motivated by a country’s own defense, but its state in a bystander spills over to third parties, since the reach of Western export controls depended on how many citizens of Kazakhstan still identified with the Russian-speaking world. Finally, our results show that media policy is a weaker lever than it appears. The language cue moves attitudes with no media exposure, so restricting the aggressor’s outlets addresses the persistence of the divide but not its activation.

6 Conclusion

Like the Brexit referendum or the COVID-19 pandemic, Russia’s full-scale invasion of Ukraine sorted people into groups, and in Kazakhstan the dividing line was language. Before February 2022, Russian and Kazakh speakers held similar views of Russia. After the invasion, their opinions of Russia diverged, and Russian-speaking regions turned to pro-Kremlin outlets. Addressing respondents in Russian on a survey’s introductory page made them more likely to justify the invasion and help with sanctions evasion, although it changed neither their information nor their incentives. Once the groups had formed, media consumption followed the new divide and kept it open through late 2023, much as group dynamics deepened the divides after Brexit and the pandemic ([Hobolt and Tilley 2026](#)).

This sorting is familiar from established democracies, but we find it in a country that is not party to the war, and there it had economic consequences beyond the country’s borders. Regions where Russian is widely spoken among ethnic Kazakhs re-exported far more sanctioned goods to Russia, so the reach of Western export controls depended on how citizens of a third country identified. Our results also qualify the view that an external war strengthens national identity. In countries under attack or under direct threat, the population tends to unite around the nation ([Dehdari and Gehring 2022](#); [Gehring 2022](#); [Abramenko et al. 2026](#)). In Kazakhstan, the same war widened a gap between language groups that had not appeared in attitudes toward Russia before 2022. The trait that divided Kazakhstan was the one Russia used to justify the war, so a neighboring power could change which division mattered in the country without having to persuade its citizens. For states that share a language with a former imperial power, the cohesion their governments build at home therefore depends partly on decisions taken abroad. Our data cannot tell which of Kazakhstan’s language policies, the

shift of schooling to Kazakh or the equal legal status of Russian, made the country more or less exposed to this divide.

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Online Appendix

A Appendix

Example for the conceptual model. Take three binary traits and set $\delta_k = 0$ when i matches the group’s prototype on trait k and $\delta_k = 1$ when she does not.

The attacked country. In Ukraine, the war makes a trait salient that carried little weight before, namely the *side* of the interstate boundary a person stands on. Suppose the prototype of R is (ethnic Russian, Russian-speaking, Russia’s side) and that of the Ukrainian nation N is (ethnic Ukrainian, Ukrainian-speaking, Ukraine’s side). Let us consider a Russian-speaking ethnic Russian resident of Kharkiv who matches R on the first two traits but stands on Ukraine’s side. Before the invasion, with equal weights, she is

$$d_{iR} = \frac{1}{3}(0) + \frac{1}{3}(0) + \frac{1}{3}(1) = \frac{1}{3}, \quad d_{iN} = \frac{1}{3}(1) + \frac{1}{3}(1) + \frac{1}{3}(0) = \frac{2}{3},$$

closer to R . When the invasion raises the weight on the interstate side to $\frac{3}{5}$ (leaving $\frac{1}{5}$ on each other trait),

$$d_{iR} = \frac{1}{5}(0) + \frac{1}{5}(0) + \frac{3}{5}(1) = \frac{3}{5}, \quad d_{iN} = \frac{1}{5}(1) + \frac{1}{5}(1) + \frac{3}{5}(0) = \frac{2}{5},$$

she flips to N . A Ukrainian-speaking ethnic Ukrainian sits at $d_{iN} = 0$ and $d_{iR} = 1$ under any weights and does not move. The reweighting therefore shifts only the Russian-leaning residents, and shifts them toward N – consolidation, largest exactly where pre-war attachment to N was weakest, as [Abramenko et al. \(2026\)](#) find.

The bystander. Kazakhstan is not a party to the war, so no “side” trait exists for its residents; the relevant traits are language, ethnicity, and culture. Suppose the prototype of the Russian-speaking world R is (ethnic Russian, Russian-speaking, Orthodox) and that of the Kazakh nation N is (ethnic Kazakh, Kazakh-speaking, Muslim). A person i who is ethnic Kazakh, Russian-speaking, and atheist, weighting the three traits equally ($w_k = 1/3$), is then

$$d_{iR} = \frac{1}{3}(1) + \frac{1}{3}(0) + \frac{1}{3}(1) = \frac{2}{3}, \quad d_{iN} = \frac{1}{3}(0) + \frac{1}{3}(1) + \frac{1}{3}(1) = \frac{2}{3},$$

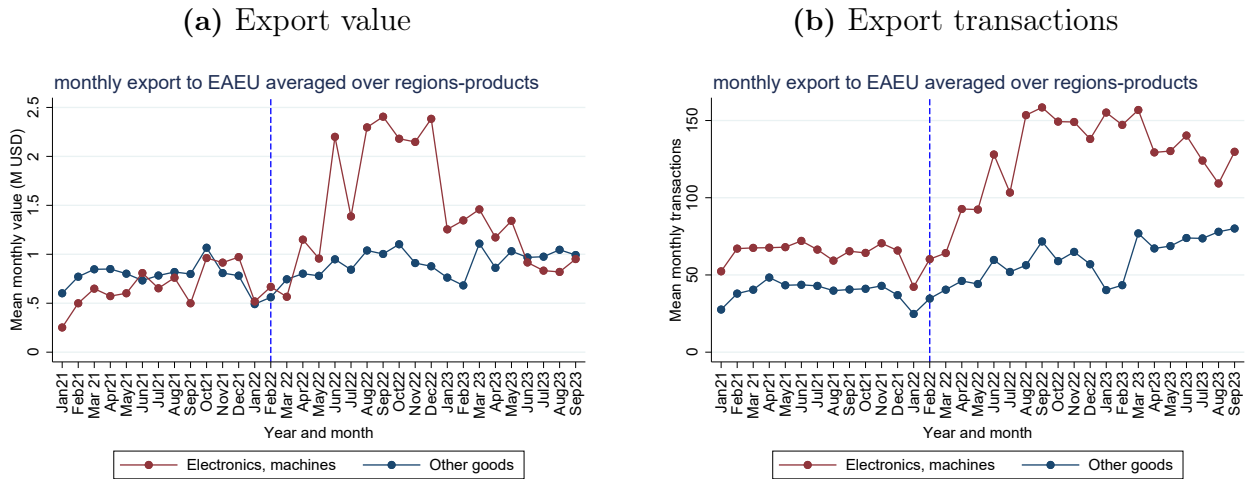
equally distant from both prototypes. Raising the weight on language alone to $1/2$ pulls her toward the group whose language she shares:

$$d_{iR} = \frac{1}{4}(1) + \frac{1}{2}(0) + \frac{1}{4}(1) = \frac{1}{2}, \quad d_{iN} = \frac{1}{4}(0) + \frac{1}{2}(1) + \frac{1}{4}(1) = \frac{3}{4},$$

so she now sits closer to R than to N , while her Kazakh-speaking neighbor with otherwise similar traits stays closer to N .

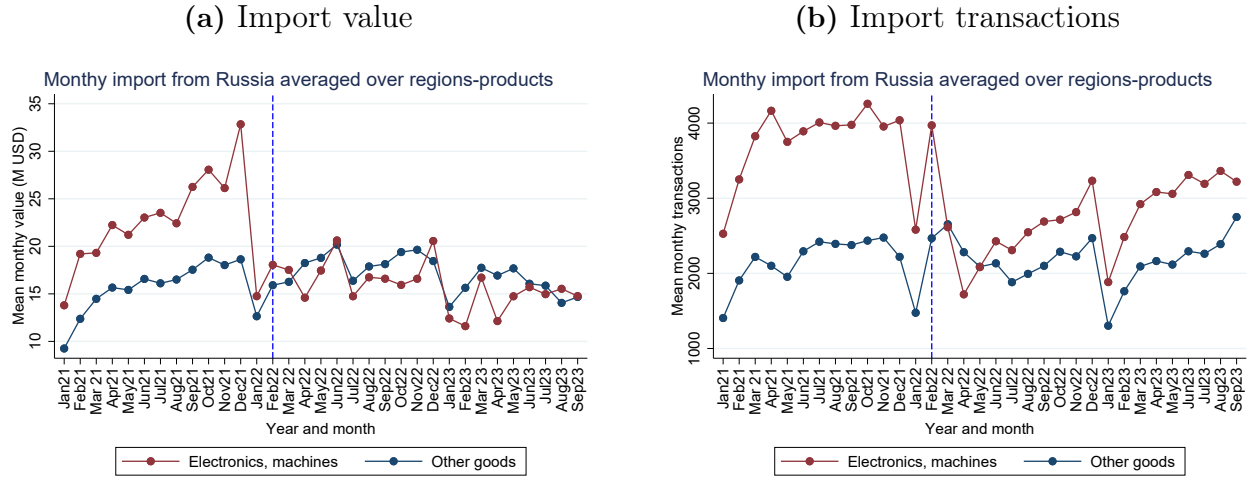
The two cases differ in one structural feature. In the attacked country the war lifts the weight on a trait every resident shares (being on the attacked side), so the reweighting pulls all residents toward N together. In the bystander the war lifts the weight on a trait residents sit on opposite ends of (language), so the same mechanism pulls them apart. Whether war consolidates or polarizes thus depends on whether the trait it makes salient unites or divides the population.

Figure A1: Export from Kazakhstan to EAEU countries excluding Russia



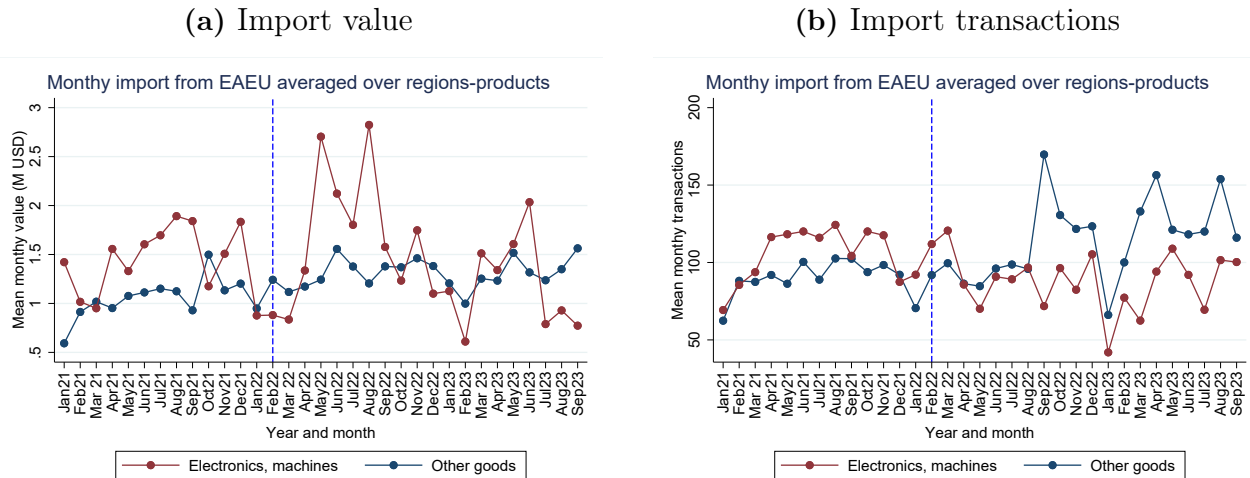
Note. The figure shows the monthly value of export in millions of USD (left) and the number of export transactions (right) of Kazakhstan to EAEU countries excluding Russia, averaged over 5 product classes and regions for 2021-2023. For more details, see Section 3.

Figure A2: Import to Kazakhstan from Russia



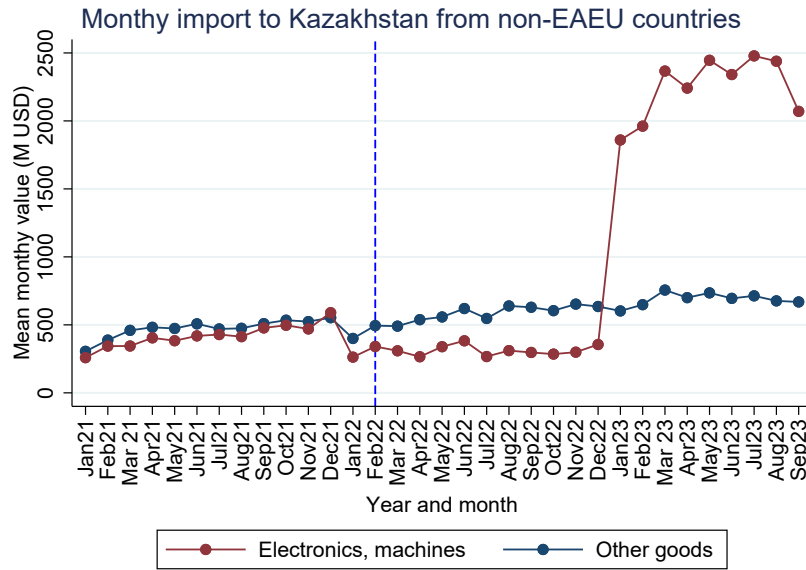
Note. The figure shows the monthly value of imports in millions of USD (left) and the number of import transactions (right) to Kazakhstan from Russia, averaged over 5 product classes and regions for 2021-2023. For more details, see Section 3.

Figure A3: Import to Kazakhstan from EAEU countries excluding Russia



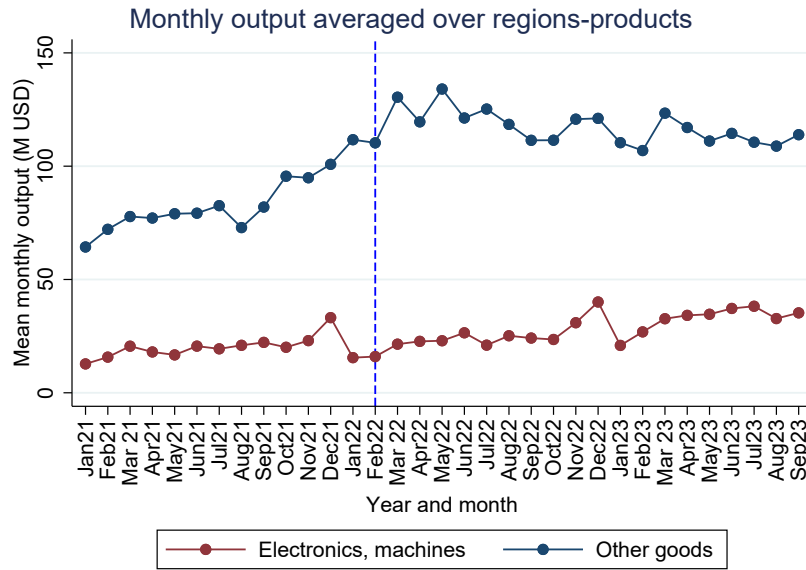
Note. The figure shows the monthly value of imports in millions of USD (left) and the number of import transactions (right) to Kazakhstan from EAEU countries excluding Russia, averaged over 5 product classes and regions for 2021-2023. For more details, see Section 3.

Figure A4: World imports to Kazakhstan



Note. The figure shows the monthly world value of imports to Kazakhstan in millions of USD, averaged across 5 product classes for 2021-2023. The data are collected from the [UN Comtrade Database](#) at the monthly-HS2 code level, covering total imports into Kazakhstan from the world.

Figure A5: The production output of Kazakhstan



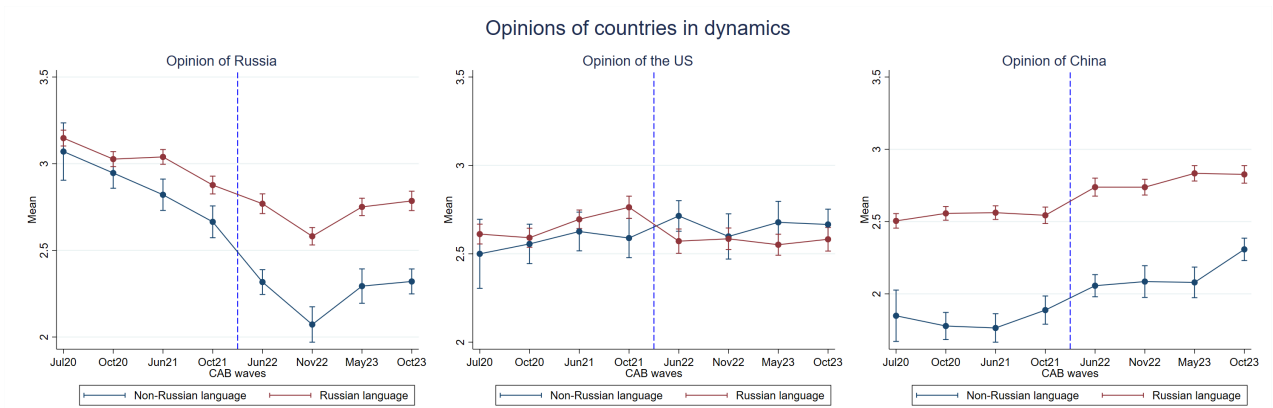
Note. The figure shows the monthly value of production, averaged over 5 product classes and regions for 2021-2023. For more details, see Section 3.

Table A1: Descriptive statistics for the war attitudes and responsibility attribution

Year	SMO is justified	War responsibility (%)		
		Russia	West	Ukraine
2022	26.3	27.5	10.8	20.8
2023	25	24.9	12.1	19.4

Note. The table uses CAB survey waves for 2022-2023. The “SMO is justified” column shows the percentage of “Completely justified” or “Somewhat justified” opinions.

Figure A6: Opinions of Russia, the US, and China by survey language over time



Note. The figure shows the mean trends in opinions across different countries by survey language (Russian or Kazakh), based on CAB data for 2020-2023.

Figure A7: Search for “War Ukraine” in the Kazakh language on Yandex

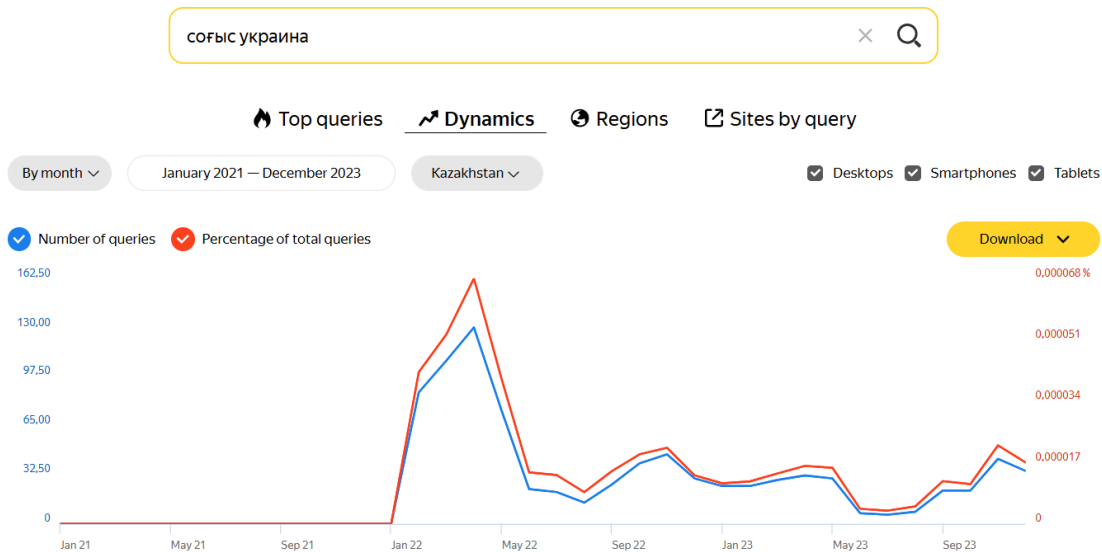


Figure A8: Search for “War Ukraine” in the Kazakh language on Google Trends

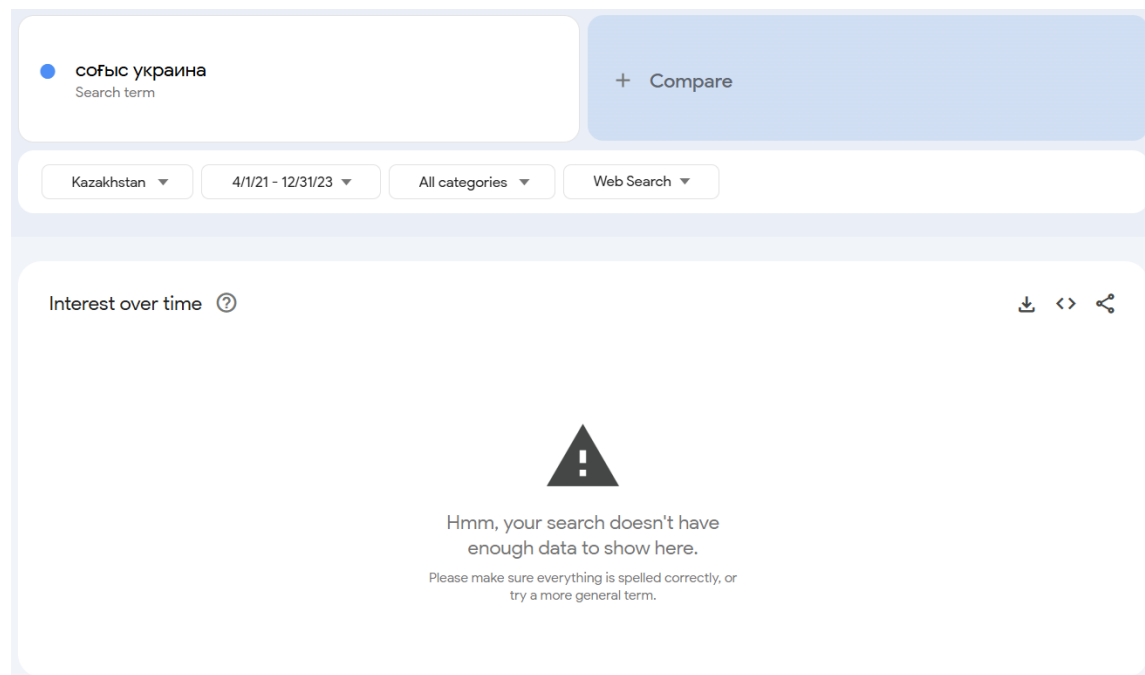


Figure A9: Search for “News” on Yandex

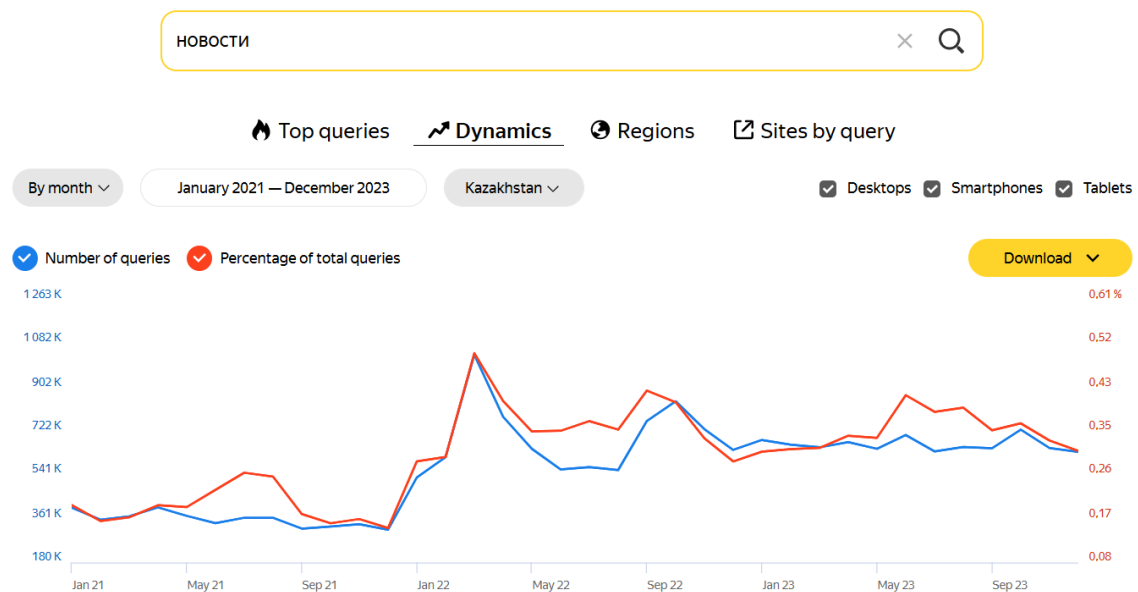


Figure A10: Search for “Yandex Dzen” on Yandex

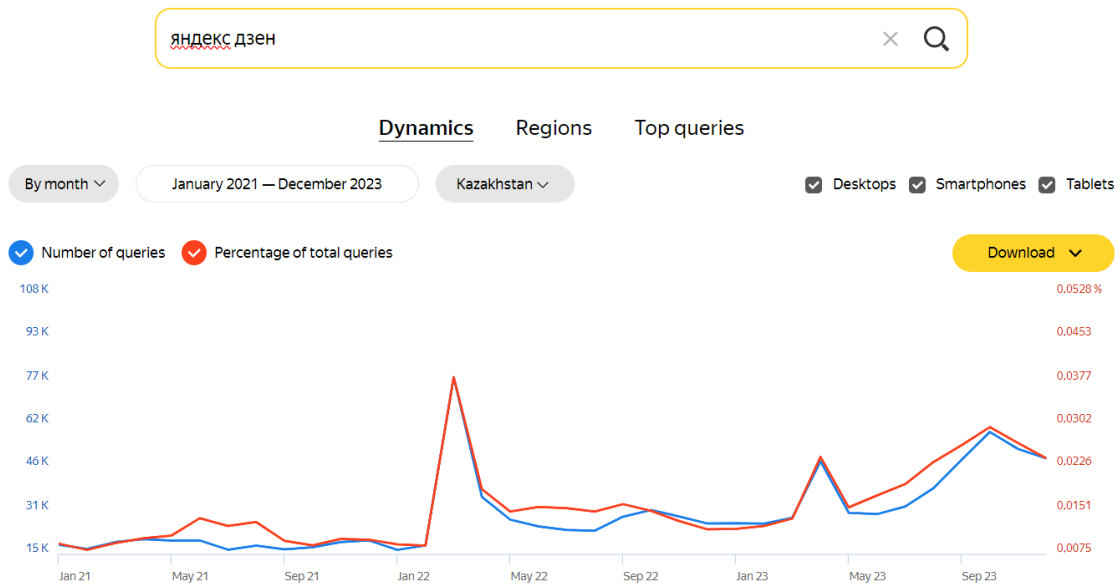


Figure A11: Search for “Yandex News” on Yandex

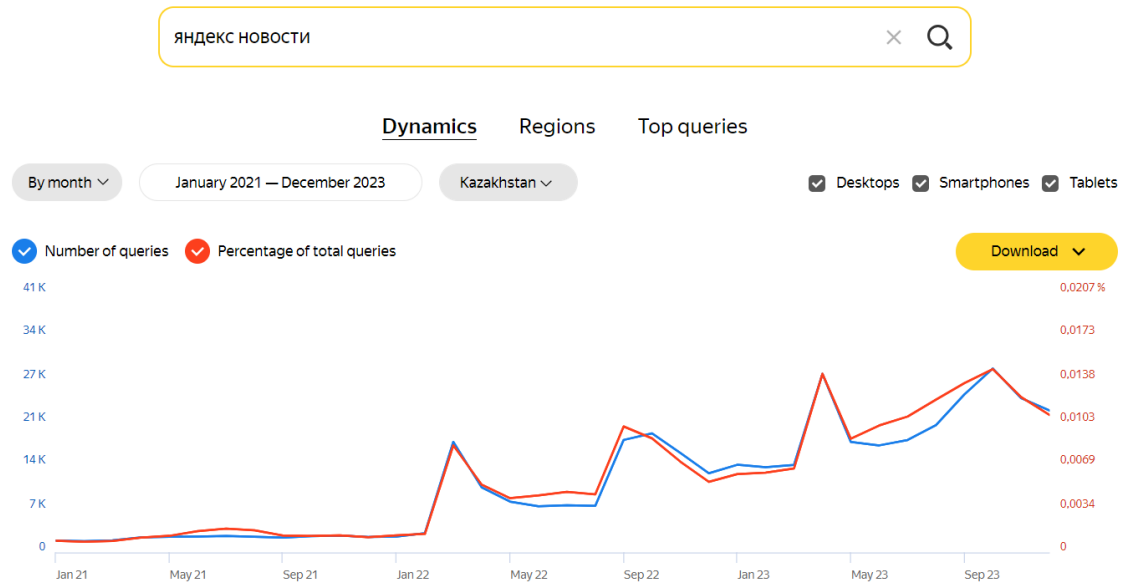
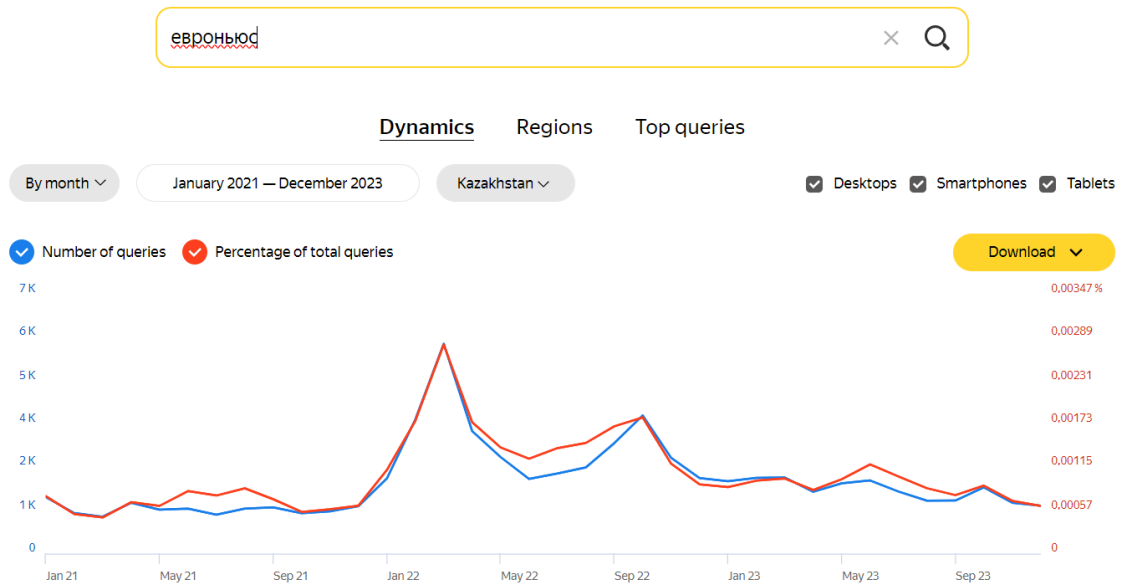
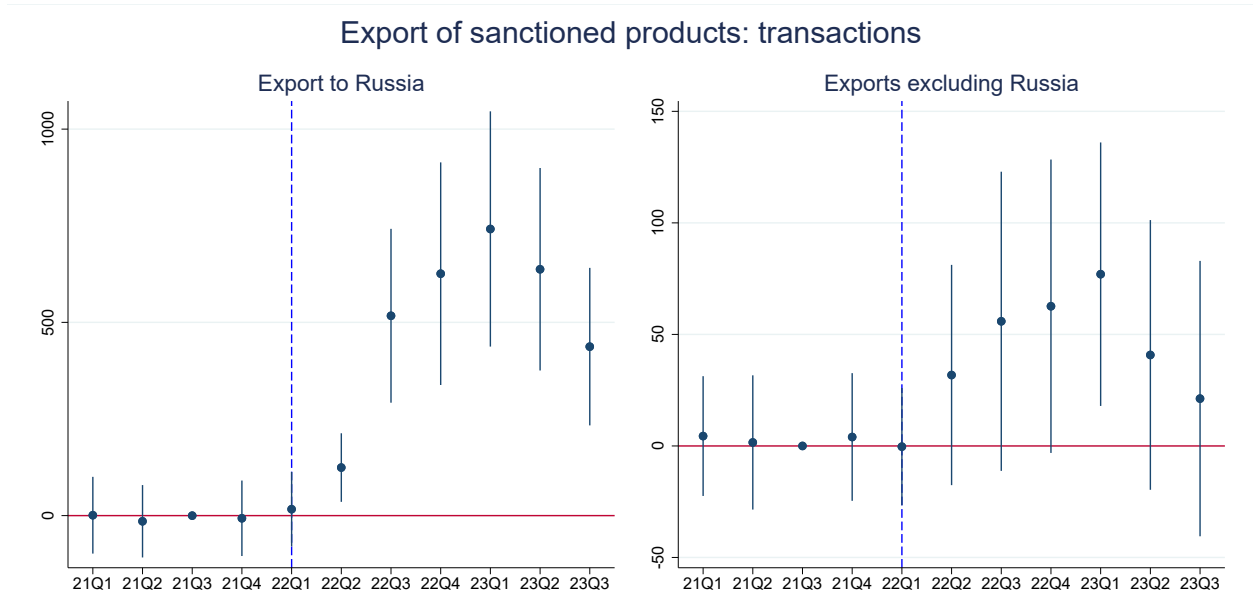


Figure A12: Search for “Euronews” on Yandex



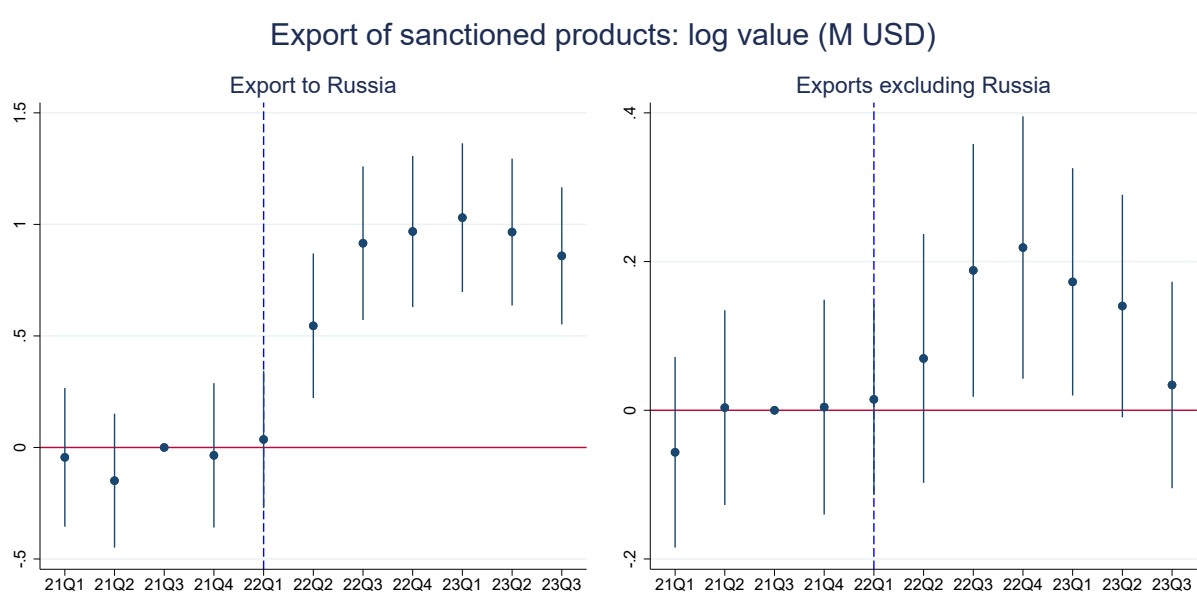
B Appendix

Figure B1: The number of export transactions growth for sanctioned products: event-study



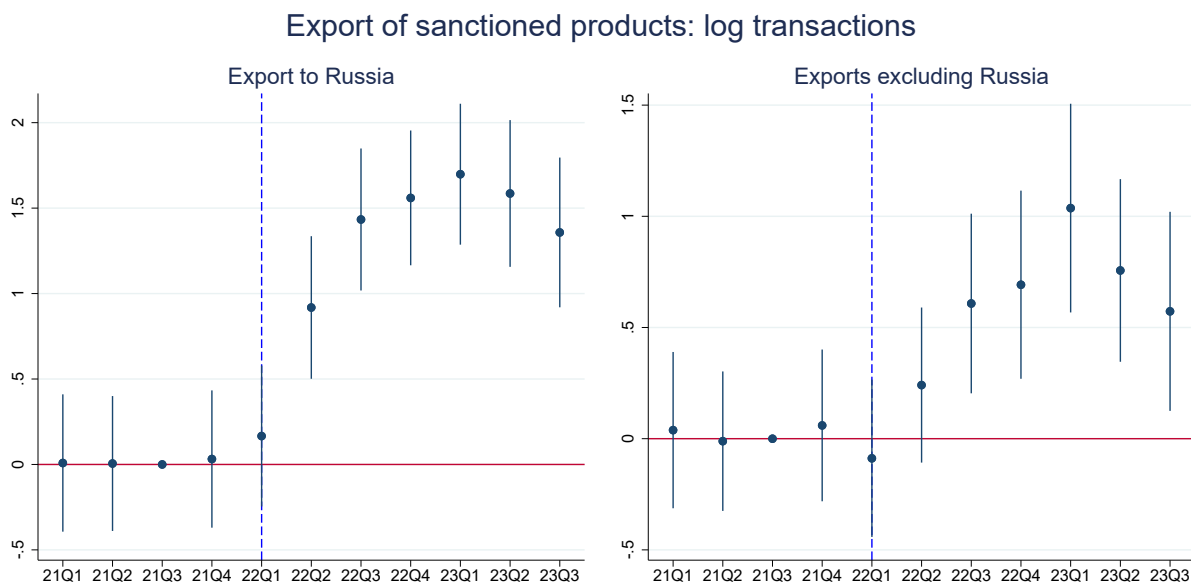
Note. The graph shows the event-study coefficients from (3). Blue dots show the coefficients β_τ . The 95% confidence intervals are calculated w.r.t. robust standard errors.

Figure B2: The log of the export value growth for sanctioned products: event-study



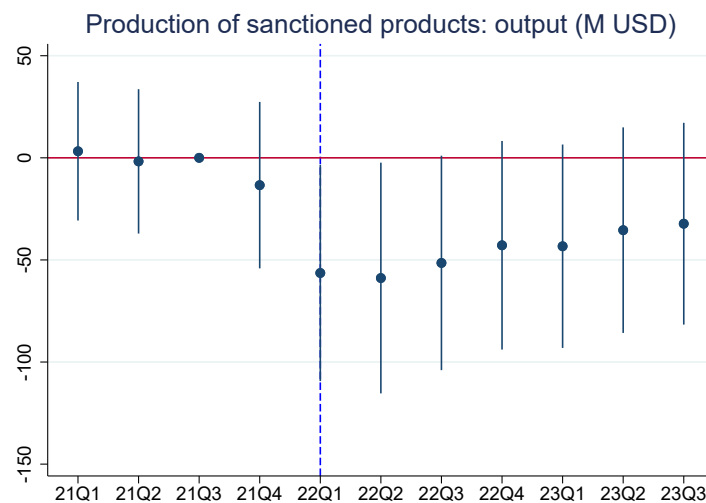
Note. The graph shows the event-study coefficients from (3). Blue dots show the coefficients β_τ . The 95% confidence intervals are calculated w.r.t. robust standard errors.

Figure B3: The log of the number of export transactions growth for sanctioned products: event-study



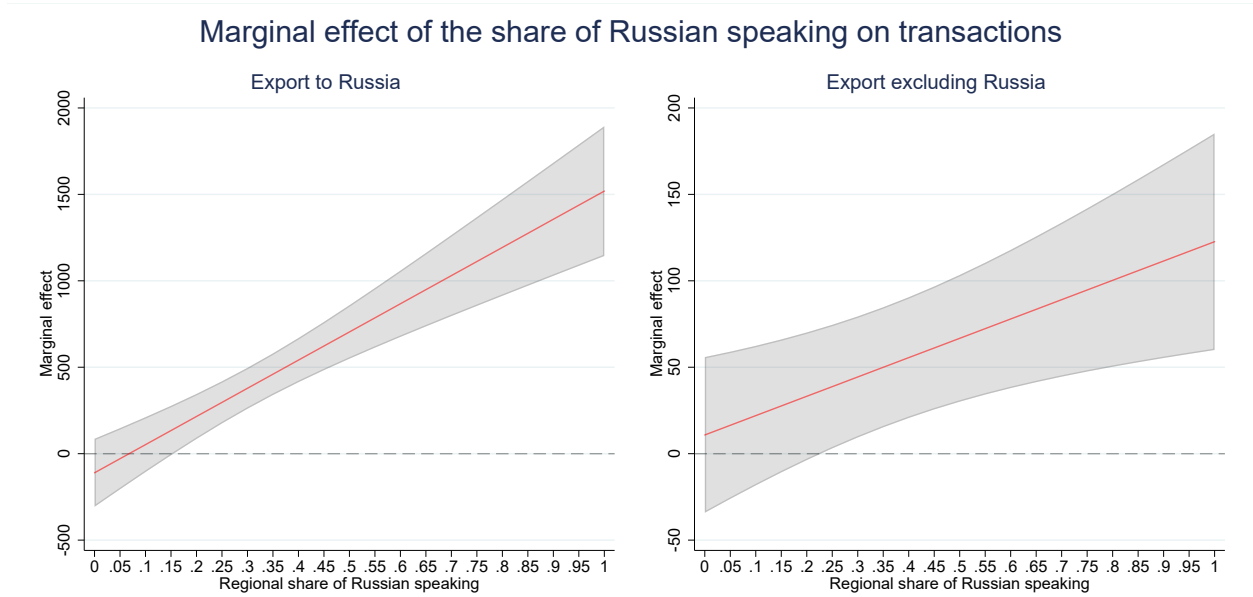
Note. The graph shows the event-study coefficients from (3). Blue dots show the coefficients β_τ . The 95% confidence intervals are calculated w.r.t. robust standard errors.

Figure B4: The output change for sanctioned products: event-study



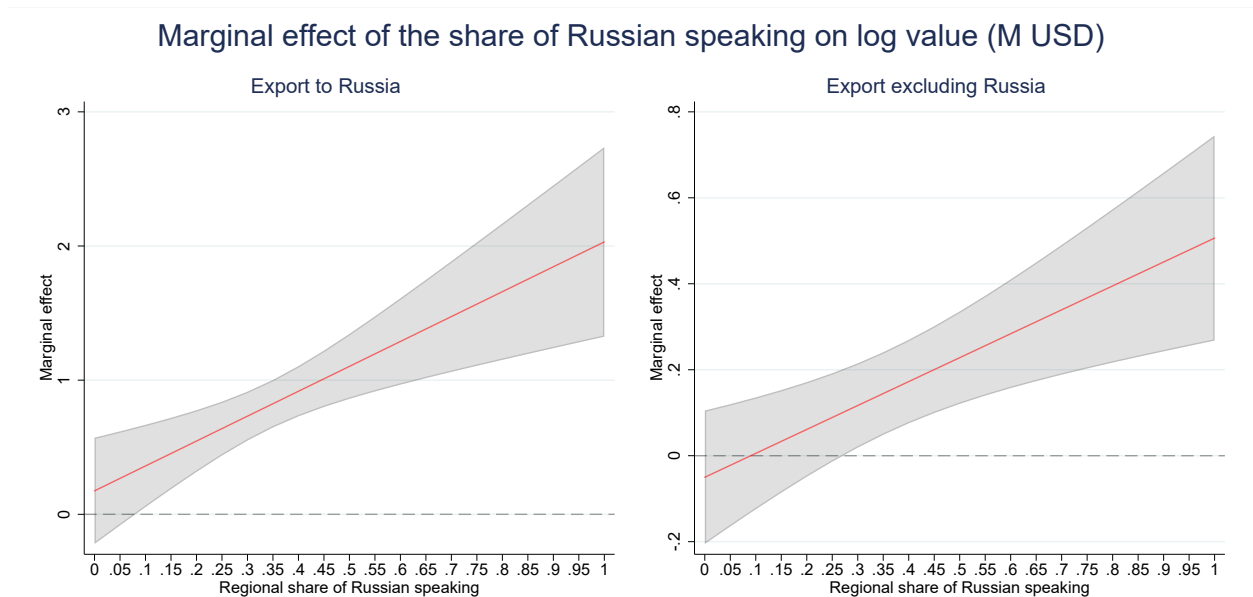
Note. The graph shows the event-study coefficients from (3). Blue dots show the coefficients β_τ . The 95% confidence intervals are calculated w.r.t. robust standard errors.

Figure B5: The number of transactions growth for sanctioned products by regional share of Russian-speaking Kazakhs



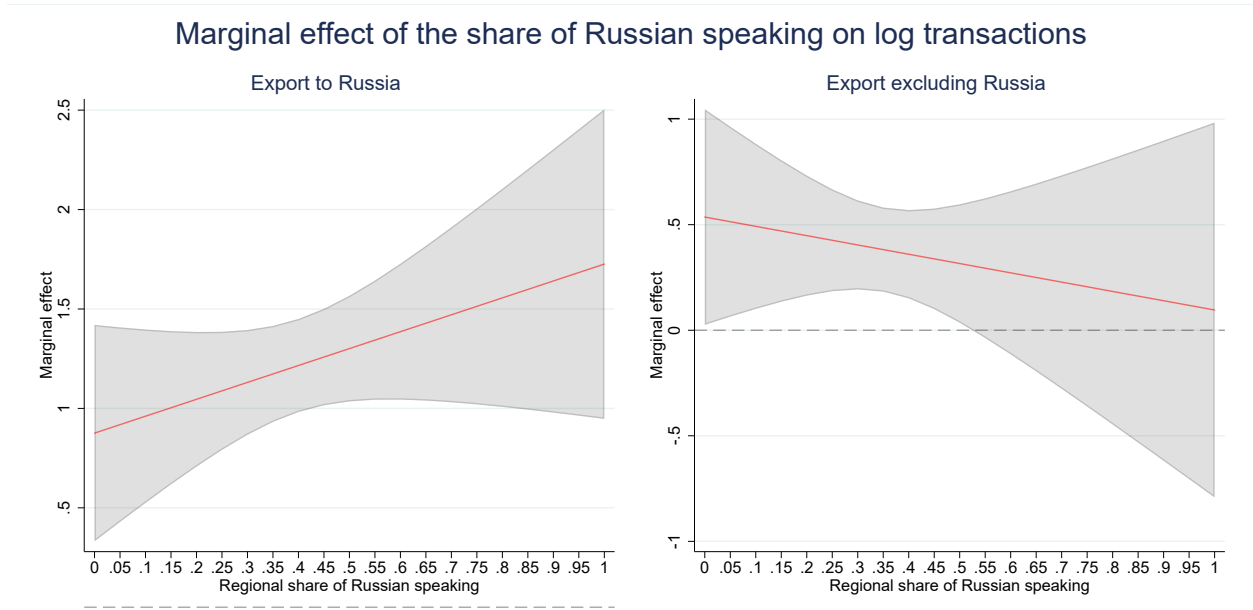
Note. The figure shows $\beta + \gamma \cdot Treat$ from the specification (4). The dependent variable is the number of transactions for a product class in a region-month. The regional share of Russian-speaking Kazakhs is based on the 2021 Census.

Figure B6: The log of the value of export growth for sanctioned products by regional share of Russian-speaking Kazakhs



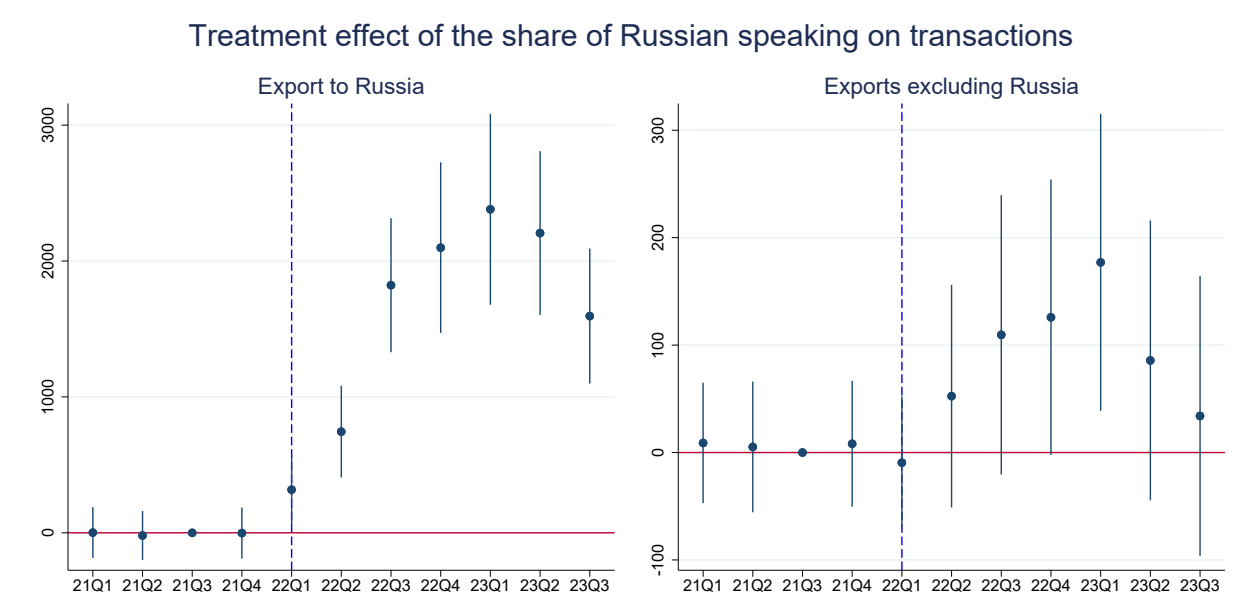
Note. The figure shows $\beta + \gamma \cdot Treat$ from the specification (4). The dependent variable is the log of export value for a product class in a region-month. The regional share of Russian-speaking Kazakhs is based on the 2021 Census.

Figure B7: The log of the number of transactions growth for sanctioned products by regional share of Russian-speaking Kazakhs



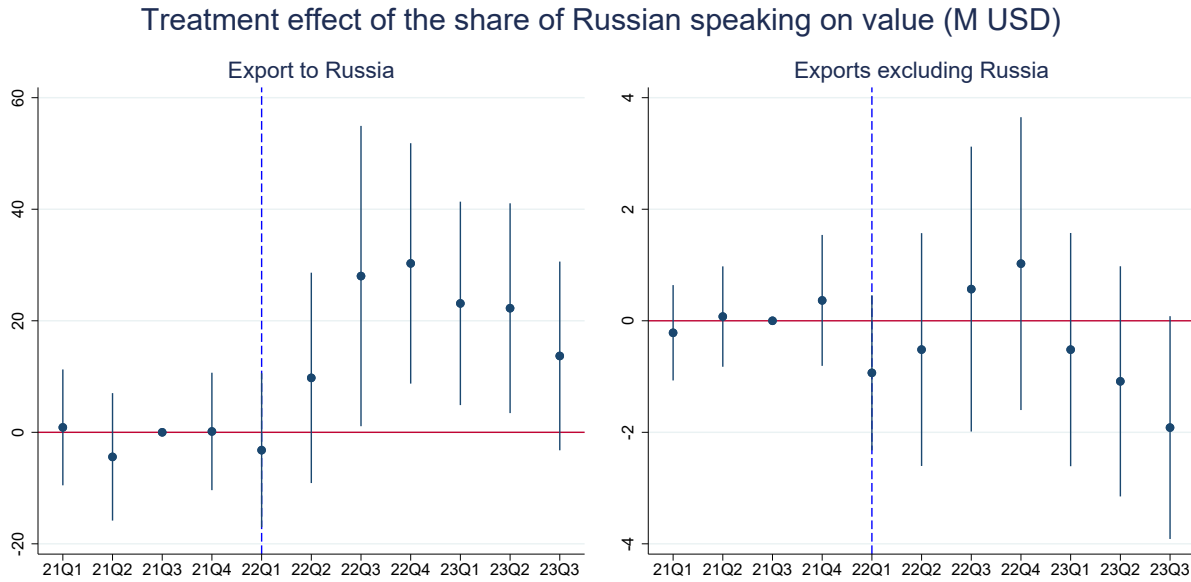
Note. The figure shows $\beta + \gamma \cdot Treat$ from the specification (4). The dependent variable is the log of the number of transactions for a product class in a region-month. The regional share of Russian-speaking Kazakhs is based on the 2021 Census.

Figure B8: Event study for the transaction growth by regional share of Russian-speaking Kazakhs



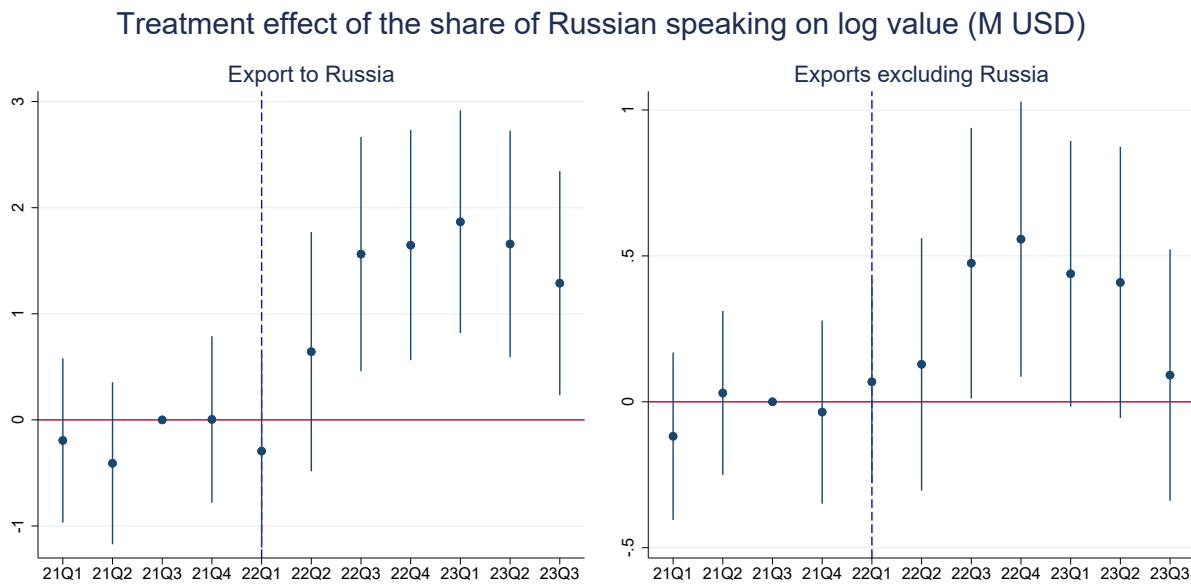
Note. The figure shows coefficients γ_τ from (5). The dependent variable is the number of transactions for a product class in a region-month.

Figure B9: Event study for the value of export growth by regional share of Russian-speaking Kazakh, net of regional share of ethnic Russians



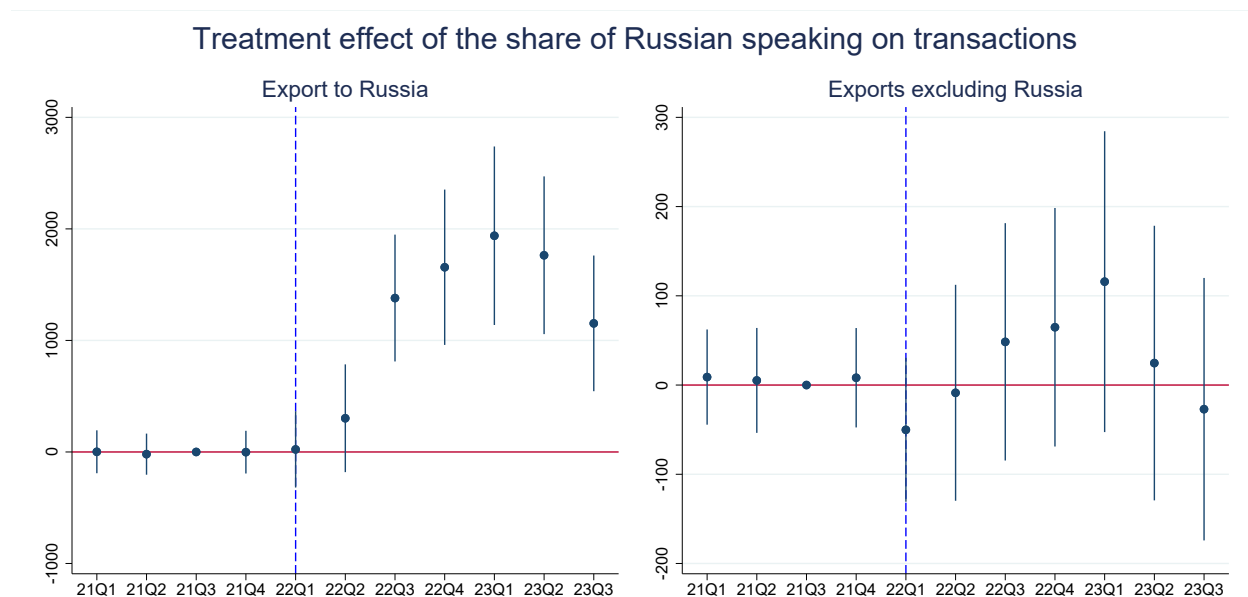
Note. The figure shows coefficients γ_τ from (6). The dependent variable is the value of export for a product class in a region-month.

Figure B10: Event study for the log of the export value growth by regional share of Russian-speaking Kazakh, net of regional share of ethnic Russians



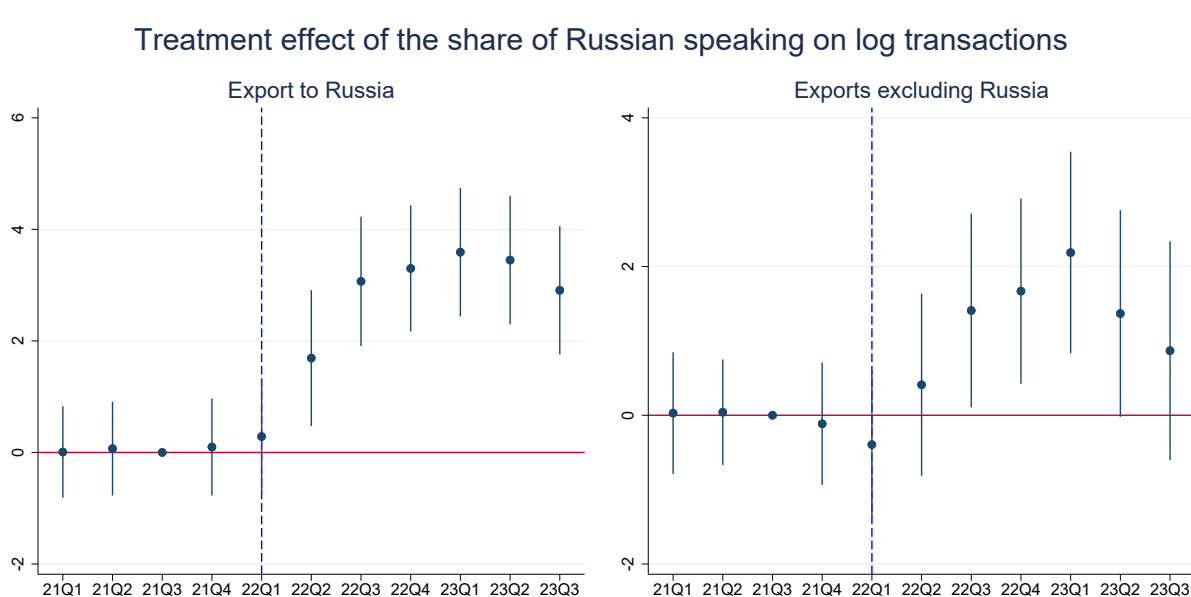
Note. The figure shows coefficients γ_τ from (6). The dependent variable is the log of export value for a product class in a region-month.

Figure B11: Event study for the transactions growth by regional share of Russian-speaking Kazakh, net of regional share of ethnic Russians



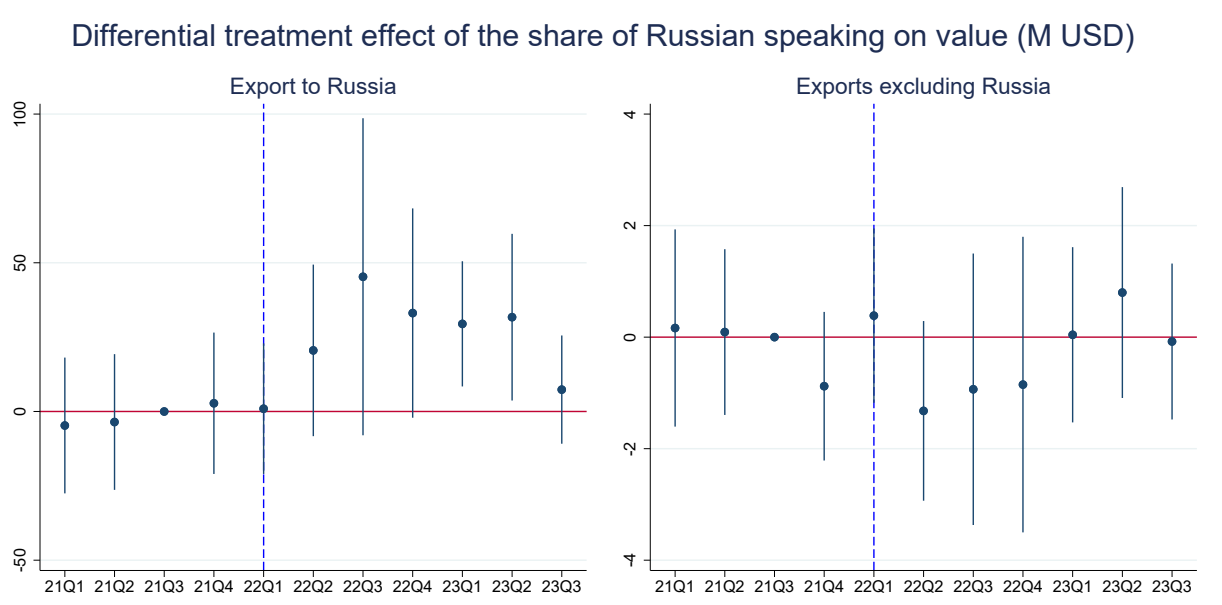
Note. The figure shows coefficients γ_τ from (6). The dependent variable is the number of transactions for a product class in a region-month.

Figure B12: Event study for the log of transaction growth by regional share of Russian-speaking Kazakh, net of regional share of ethnic Russians



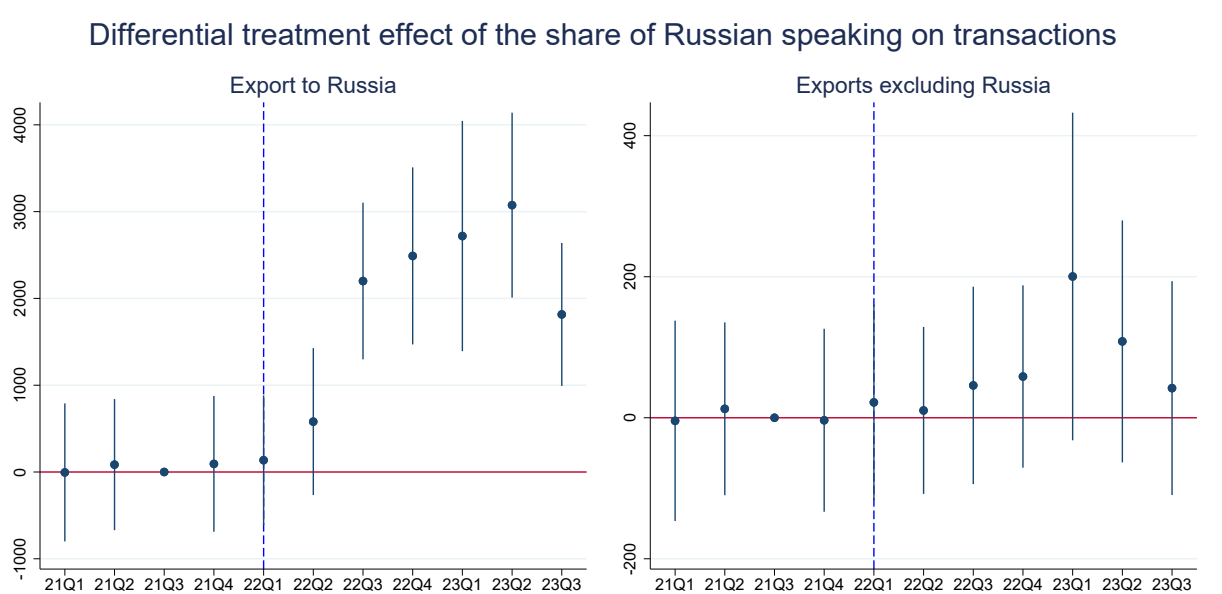
Note. The figure shows coefficients γ_τ from (6). The dependent variable is the log of the transaction number for a product class in a region-month.

Figure B13: The event study of the effect of the Russian language on export value growth



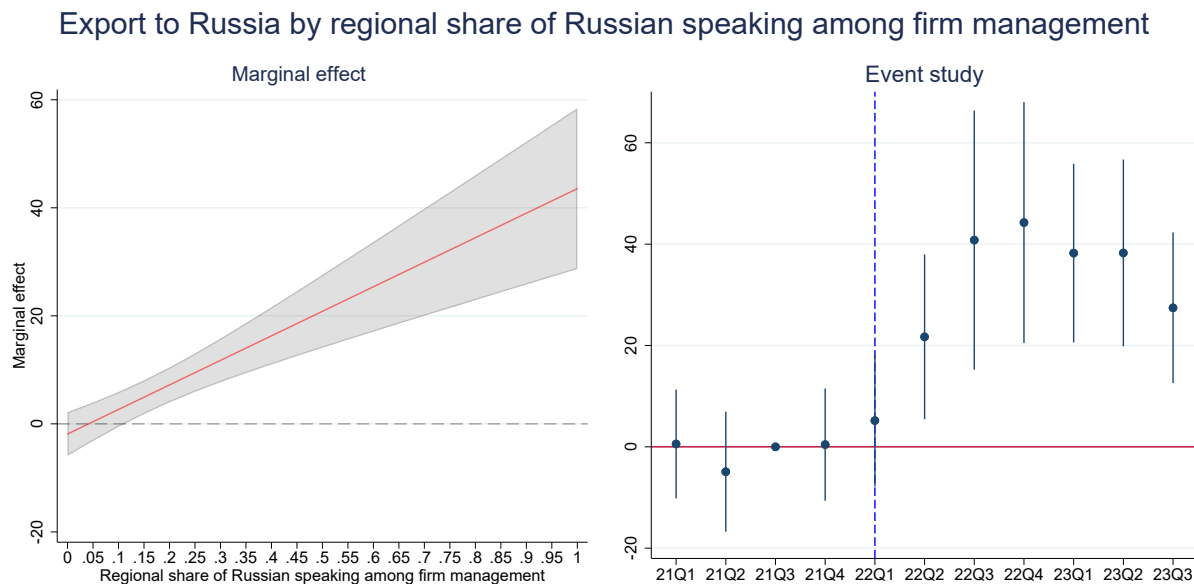
Note. The figure shows the event-study results from the DID specification (7) (i.e., γ_τ). The dependent variable is the export value for sanctioned products in a region-month. The regional share of Russian-speaking Kazakhs is based on the 2021 Census.

Figure B14: The event study of the effect of Russian language on transaction growth



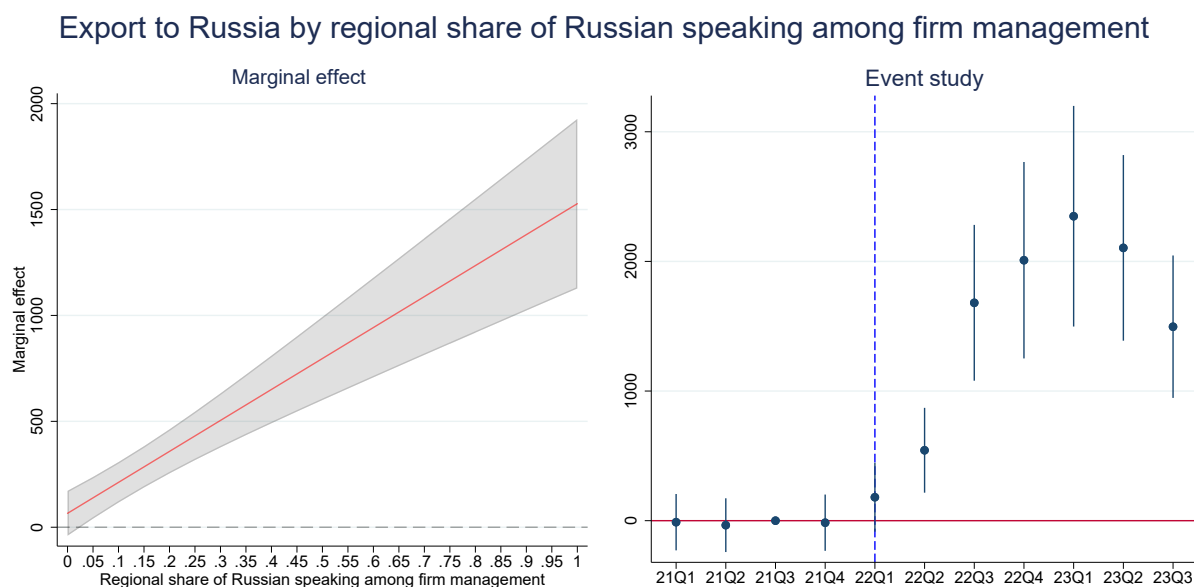
Note. The figure shows the event-study results from the DID specification (7) (i.e., γ_τ). The dependent variable is the number of transactions for sanctioned products in a region-month. The regional share of Russian-speaking Kazakhs is based on the 2021 Census.

Figure B15: The effect of the Russian language among firm management on export value growth to Russia (M USD)



Note. The figure shows the marginal effect $\beta + \gamma \cdot Treat$ from the specification (4) (left) and the event-study coefficients γ_τ from (5) (right). Regional treatment exposure is calculated among ethnic Kazakh firm managers of CAB 2020-2021 waves as the regional share of speaking Russian at home as the main language. The dependent variable is the export value for a product class in a region-month.

Figure B16: The effect of the Russian language among firm management on export transactions to Russia



Note. The figure shows the marginal effect $\beta + \gamma \cdot Treat$ from the specification (4) (left) and the event-study coefficients γ_τ from (5) (right). Regional treatment exposure is calculated among ethnic Kazakh firm managers of CAB 2020-2021 waves as the regional share of speaking Russian at home as the main language. The dependent variable is the number of export transactions for a product class in a region-month.

C Appendix

Table C1: Summary statistics for the online survey

	(1)		
	count	mean	sd
Ethnic Russian	5025	0.24	0.43
Female	5025	0.51	0.50
Age	5025	42.65	16.13
Number of children	5025	1.26	1.46
Higher education	5025	0.31	0.46
Work full-time or part-time	5025	0.61	0.49
Household welfare (scale 1-5)	5025	2.85	1.14
Internet usage (scale 1-6)	5025	4.47	1.80
Rural area	5025	0.16	0.36
Intro page in Russian	5025	0.49	0.50
Survey in Russian	5025	0.53	0.50
Russian at home	5025	0.38	0.48
Russian with friends	5025	0.41	0.49
News in Russian	5025	0.45	0.50
Sanctions evasion: justified	5025	0.30	0.46
Invasion/SMO: justified	5025	0.26	0.44
Opinion of Russia (scale 1-5)	4334	3.36	1.02
Opinion of China (scale 1-5)	4327	3.28	0.99
Opinion of the US (scale 1-5)	4366	3.34	1.09
Government trust (scale 1-4)	4340	2.99	0.94
Protest participation: justified	5025	0.32	0.46
Work $\geq$ 50 h.: justified	5025	0.31	0.46

Note. The table summarizes the characteristics of the participants in the online survey, including demographics, language use, socioeconomic status, and attitudes toward Russian and non-Russian affairs.

Table C2: Balance statistics for the language experiment in an online survey

Variable	(1)	(2)	(3)	(4)
	Randomized introductory page Kazakh	Russian	Difference	Cohen's d
Ethnic Russian	0.238 (0.426)	0.252 (0.434)	0.014 (0.012)	0.023
Female	0.514 (0.500)	0.511 (0.500)	-0.003 (0.014)	-0.004
Age	42.651 (16.276)	42.659 (15.975)	0.008 (0.455)	0.000
Number of children	1.295 (1.480)	1.222 (1.440)	-0.073* (0.041)	-0.035
Higher education	0.306 (0.461)	0.309 (0.462)	0.002 (0.013)	0.004
Work full-time or part-time	0.606 (0.489)	0.619 (0.486)	0.013 (0.014)	0.019
Household welfare (scale 1-5)	2.859 (1.142)	2.845 (1.129)	-0.014 (0.032)	-0.009
Internet usage (scale 1-6)	4.492 (1.789)	4.438 (1.805)	-0.054 (0.051)	-0.021
Rural area	0.155 (0.362)	0.157 (0.364)	0.002 (0.010)	0.004
Observations	2,542	2,483	5,025	
Completion rate, all ad clicks	0.181 (0.385)	0.178 (0.382)	-0.003 (0.005)	-0.007
Completion rate, eligible ad clicks	0.235 (0.424)	0.240 (0.427)	0.006 (0.006)	0.013
Ad clicks, all	14,066	13,955	28,021	
Ad clicks, eligible	10,833	10,336	21,169	

Note. The table shows the balance statistics of the randomly assigned introductory page in the online survey for demographics. The upper panel covers the 5,025 respondents who completed the survey and passed all quality checks. The lower panel covers all users who clicked the survey ad and were randomly assigned an introductory page. Completion rate equals 1 if the user completed the survey and passed all quality checks. Eligible ad clicks exclude users screened out for being under 18, for answering “Other” to the region question, or for an IP address or payment phone number that was already used or previously blocked. Standard deviations are in parentheses in Columns 1 and 2, and standard errors of the difference in Column 3.

Table C3: Russian introductory page and attitudes toward non-Russian affairs

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Opinion of China		Opinion of the US		Government trust		Protest participation		Work $\geq$ 50 h.	
Intro page in Russian	0.051* (0.030)	0.040 (0.029)	0.0025 (0.033)	0.0044 (0.029)	0.047* (0.028)	0.041 (0.027)	-0.0021 (0.013)	-0.0015 (0.012)	-0.0038 (0.013)	-0.0026 (0.012)
Constant	3.25*** (0.021)	3.71*** (0.11)	3.34*** (0.023)	3.90*** (0.11)	2.97*** (0.020)	3.51*** (0.094)	0.32*** (0.0092)	0.55*** (0.048)	0.32*** (0.0092)	0.39*** (0.046)
Observations	4,327	4,327	4,366	4,366	4,340	4,340	5,025	5,025	5,025	5,025
R-squared	0.001	0.080	0.000	0.219	0.001	0.144	0.000	0.112	0.000	0.107
Demographics	N	Y	N	Y	N	Y	N	Y	N	Y
Region FE	N	Y	N	Y	N	Y	N	Y	N	Y
Estimat.	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The table shows the reduced form for the effect of randomized language of the introductory page on attitudes toward non-Russian affairs. The dependent scale variables are the opinion of China (Columns 1-2) and the United States (Columns 3-4), ranging from 1 – “Very negative” to 5 – “Very positive”. The dependent scale variable in Columns 5-6 is trust in the government, ranging from 1 – “Cannot be trusted at all” to 4 – “Can be trusted completely”. In Columns 7-8, the dependent binary variable equals 1 if protest participation for political changes is “justified”. In Columns 9-10, the dependent binary variable equals 1 if working more than 50 hours per week is “justified”. Odd columns do not have controls. Even columns include additional controls: ethnicity, gender, age, number of children, education, occupation, household welfare, internet use intensity, a rural binary indicator, and regional fixed effects. Robust standard errors are in parentheses.

Table C4: The effect of Russian survey language on attitudes toward non-Russian affairs

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Opinion of China		Opinion of the US		Government trust		Protest participation		Work ≥ 50 h.	
Survey in Russian	0.21* (0.13)	0.17 (0.13)	0.011 (0.14)	0.019 (0.13)	0.19 (0.12)	0.17 (0.11)	-0.0095 (0.058)	-0.0068 (0.058)	-0.017 (0.058)	-0.012 (0.058)
Constant	3.17*** (0.066)	3.68*** (0.12)	3.33*** (0.070)	3.89*** (0.12)	2.90*** (0.058)	3.48*** (0.10)	0.32*** (0.032)	0.55*** (0.051)	0.32*** (0.032)	0.39*** (0.049)
Observations	4,327	4,327	4,366	4,366	4,340	4,340	5,025	5,025	5,025	5,025
Demographics	N	Y	N	Y	N	Y	N	Y	N	Y
Region FE	N	Y	N	Y	N	Y	N	Y	N	Y
Estimat.	IV	IV	IV	IV	IV	IV	IV	IV	IV	IV
Kleibergen-Paap Wald rk F	261.9	383.9	270.2	387.9	285.2	407.7	269.3	408.7	269.3	408.7

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The table shows the IV results of the effect of completing the survey in Russian on attitudes toward non-Russian affairs. The dependent variables (and control variables in the even columns) are the same as in Table C3. The heteroskedasticity-robust F-statistic shows that the instrument is strong (≥ 20) for standard inference (Andrews et al. 2019). Robust standard errors are in parentheses.

Table C5: The effect of Russian survey language on Russian language use

VARIABLES	(1) Russian at home	(2) Russian with friends	(3) Russian with friends	(4) Russian with friends	(5) News in Russian	(6) News in Russian
Survey in Russian	0.055 (0.059)	0.010 (0.042)	0.14** (0.057)	0.11** (0.043)	0.14** (0.057)	0.11** (0.046)
Constant	0.35*** (0.032)	0.15*** (0.038)	0.34*** (0.031)	0.055 (0.039)	0.37*** (0.031)	0.052 (0.042)
Observations	5,025	5,025	5,025	5,025	5,025	5,025
Demographics	N	Y	N	Y	N	Y
Region FE	N	Y	N	Y	N	Y
Estimat.	IV	IV	IV	IV	IV	IV
Kleibergen-Paap Wald rk F	269.3	408.7	269.3	408.7	269.3	408.7

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The table shows the IV results for the effect of completing the survey in Russian on Russian-language use across different circumstances. The heteroskedasticity-robust F-statistic shows that the instrument is strong (≥ 20) for standard inference ([Andrews et al. 2019](#)). Robust standard errors are in parentheses.

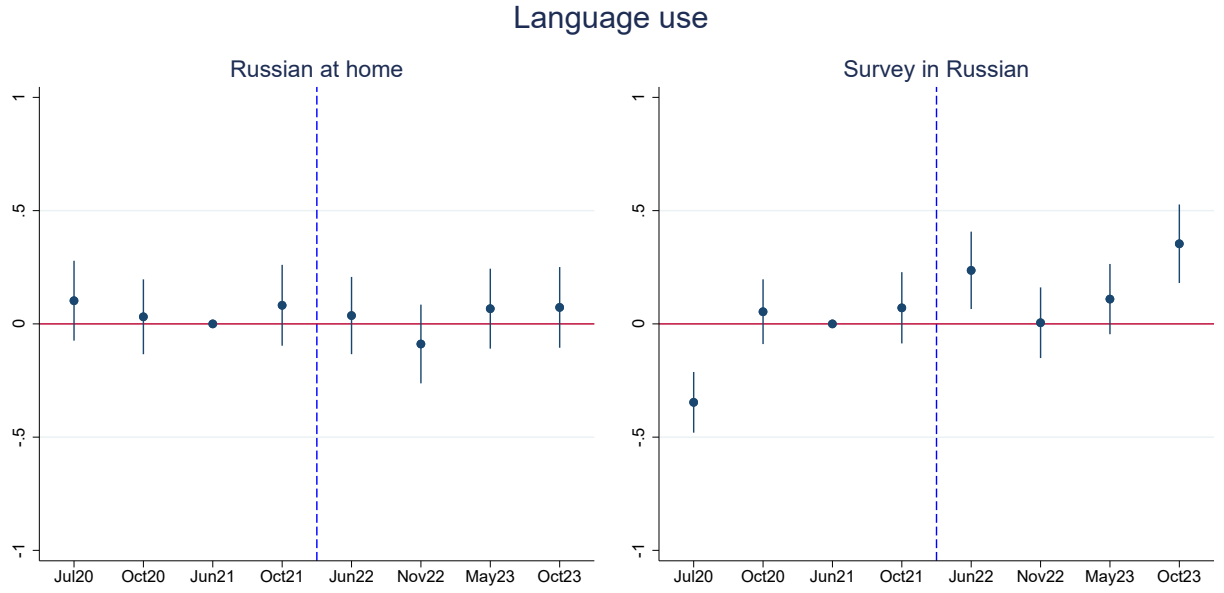
D Appendix

Table D1: Russian language use

VARIABLES	(1) Russian at home	(2)	(3) Survey in Russian	(4)
Treatment exposure (α)	0.62*** (0.033)		0.41*** (0.027)	
Post-invasion (β)	-0.0033 (0.019)		-0.18*** (0.020)	
Treatment exposure x Post-invasion (γ)	-0.023 (0.047)	-0.028 (0.047)	0.21*** (0.044)	0.22*** (0.043)
Constant	0.52*** (0.019)	0.75*** (0.038)	0.78*** (0.017)	0.97*** (0.040)
Observations	13,020	13,020	13,020	13,020
R-squared	0.490	0.512	0.228	0.291
Time FE	N	Y	N	Y
Region FE	N	Y	N	Y
Mean DV	.461	.461	.765	.765

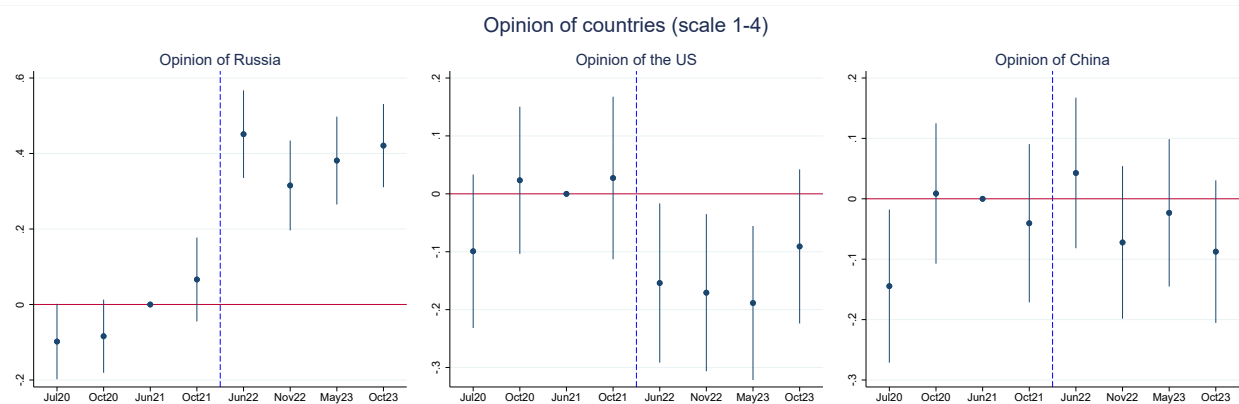
Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The odd columns show results of regression (11), even columns – (12) for binary dependent variables: “Russian at home” (Columns 1-2), “Survey in Russian” (Columns 3-4). The treatment exposure is the regional share of Russian-speaking Kazakhs from the 2021 Census. The set of control variables includes gender, five age categories, a binary indicator for higher education, five categories of household size, a binary indicator for urban settlement, Russian and Kazakh binary indicators of ethnicity, as well as the interviewer’s assessment of the respondent’s survey comprehension and comfort with answering the survey. Robust standard errors are in parentheses.

Figure D1: Event-study for Russian language use.



Note. The figure shows event-study coefficients γ_t for the DID specification (12) and dependent variables: “Russian at home” (left), “Survey in Russian” (right).

Figure D2: Event-study for Opinions of Russia, the US, and China by individual Russian language use at home



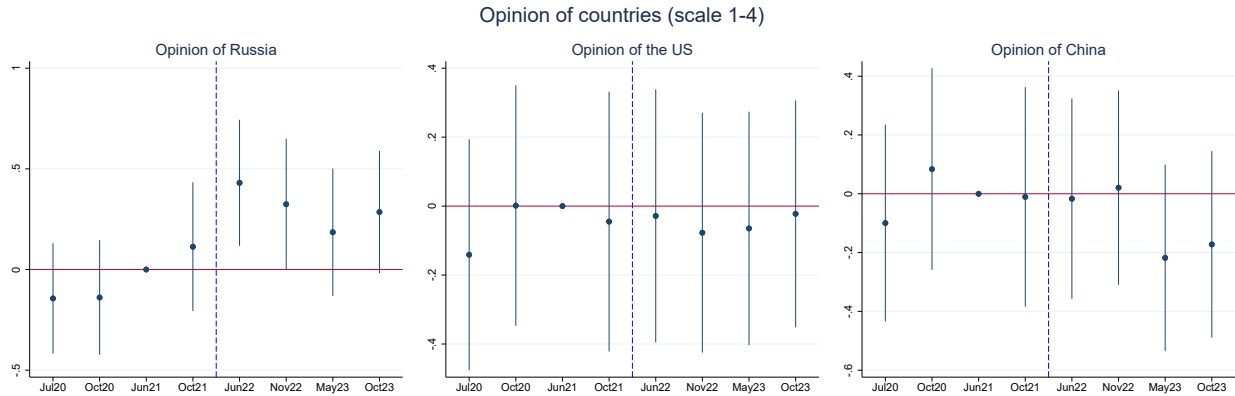
Note. The figure shows event-study coefficients γ_t for the DID specification (13) and dependent variables: “Opinion of Russia” (left), “Opinion of the US” (middle), “Opinion of China” (right). All CAB 2020-2023 respondents are in the sample.

Table D2: Opinion of Russia, the US, and China by individual Russian language use at home

VARIABLES	(1) Opinion of Russia	(2) Opinion of Russia	(3) Opinion of the US	(4) Opinion of the US	(5) Opinion of China	(6) Opinion of China
Russian at home (α)	-0.088*** (0.024)	-0.12* (0.064)	0.088*** (0.030)	-0.052 (0.073)	0.37*** (0.029)	0.32*** (0.076)
Russian at home x Post-invasion (γ)	0.42*** (0.030)	0.35*** (0.081)	-0.14*** (0.036)	-0.0017 (0.091)	0.0021 (0.033)	-0.093 (0.088)
Constant	3.29*** (0.098)	3.30*** (0.26)	2.64*** (0.11)	2.69*** (0.33)	2.49*** (0.10)	2.36*** (0.30)
Observations	11,934	1,766	10,759	1,644	11,497	1,704
R-squared	0.151	0.162	0.118	0.121	0.149	0.140
Time FE	Y	Y	Y	Y	Y	Y
Region FE	Y	Y	Y	Y	Y	Y
Mean DV	2.808	2.704	2.625	2.708	2.498	2.624
Sample	All	Managers	All	Managers	All	Managers

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The table shows OLS estimates for (13). Odd columns include all CAB 2020-2023 respondents; even columns include only CAB 2020-2023 firm managers. A person is defined as a firm manager if she declared one of the following as a primary occupation: “Business owner/Entrepreneur”, “Director/Head of company or organization”, “Head of department/Middle manager”. The dependent variables are: “Opinion of Russia” (Columns 1-2), “Opinion of the US” (Columns 3-4), “Opinion of China” (Columns 5-6) on a scale from 1 to 4. The binary treatment exposure is 1 if a respondent speaks Russian at home as the main language. The set of control variables includes gender, five age categories, a binary indicator for higher education, five categories of household size, a binary indicator for urban settlement, Russian and Kazakh binary indicators of ethnicity, as well as the interviewer’s assessment of the respondent’s survey comprehension and comfort with answering the survey. Robust standard errors are in parentheses.

Figure D3: Event-study for Opinions of Russia, the US, and China by individual Russian language use at home among firm managers



Note. The figure shows event-study coefficients γ_t for the DID specification (13) and dependent variables: “Opinion of Russia” (left), “Opinion of the US” (middle), “Opinion of China” (right). CAB 2020-2023 firm managers are in the sample. A person is defined as a firm manager if she declared one of the following as a primary occupation: “Business owner/Entrepreneur”, “Director/Head of company or organization”, “Head of department/Middle manager”.

D.1 Language Habits and Justification of the War in Ukraine

In this section, we examine the relationship between language spoken at home and attitudes toward the war in Ukraine. We use four CAB survey waves conducted after February 2022 and consider the following two-way fixed-effects linear specification:

$$y_{irt} = \alpha L_{irt} + \mathbf{X}_{irt}\beta + \kappa_r + \lambda_t + \epsilon_{irt} \quad (15)$$

The subscript i denotes a respondent from region r and wave t . The binary variable $L_{irt} = 1$ if a respondent speaks Russian as their daily language at home, and 0 otherwise. The vector $\mathbf{X}_{irt}$ includes the following set of control variables: gender, five age categories, a dummy variable for education at the university level or higher, five categories of household size, dummy variables for urban/rural settlement and ethnicity (Russian and Kazakh, compared to others), as well as the interviewer’s assessment of the respondent’s survey comprehension and comfort with answering the survey. Note that ethnicity data allow us to separate language habits from ethnic background. The variables κ_r and λ_t are region and survey-wave fixed effects, respectively. As dependent variables y_{irt} , we consider answers to the following two questions about the war in Ukraine: (i) a binary variable indicating whether SMO is justified and (ii) who is mainly responsible for the war (Russia, Ukraine, or the West). Our primary focus is on the coefficient

α , which shows the association between attitudes toward the war and language habits, while controlling for socio-demographics and ethnicity.

Columns 1-2 of Table D3 report results for the binary measure “SMO is justified” for the full sample (odd columns) and the subsample of firm managers (even columns). We find that Russian-speaking respondents are 11 p.p. more likely to agree that the SMO is justified, which is equivalent to 42% when referring to the sample average. For the firm manager subsample, the result is similar – Russian speakers are 9 p.p. more likely to justify the war. Columns 3-8 of Table D3 report results for a set of binary variables that assign responsibility for the war to three possible actors, Russia, Ukraine, or the West, as the dependent variable. In general, Russian-speaking respondents are 5.7 p.p. less likely to blame Russia as the main culprit, equivalent to 21.7% at the sample average (column 3), though the association is insignificant for firm managers (column 4). Russian speakers are not more likely to assign responsibility for the war to Ukraine (columns 5-6). However, Russian speakers 8.7 p.p. more likely to blame the West, which is equivalent to 76% when referring to the sample average (column 7). This association is also strong for the subsample of firm managers (column 8).

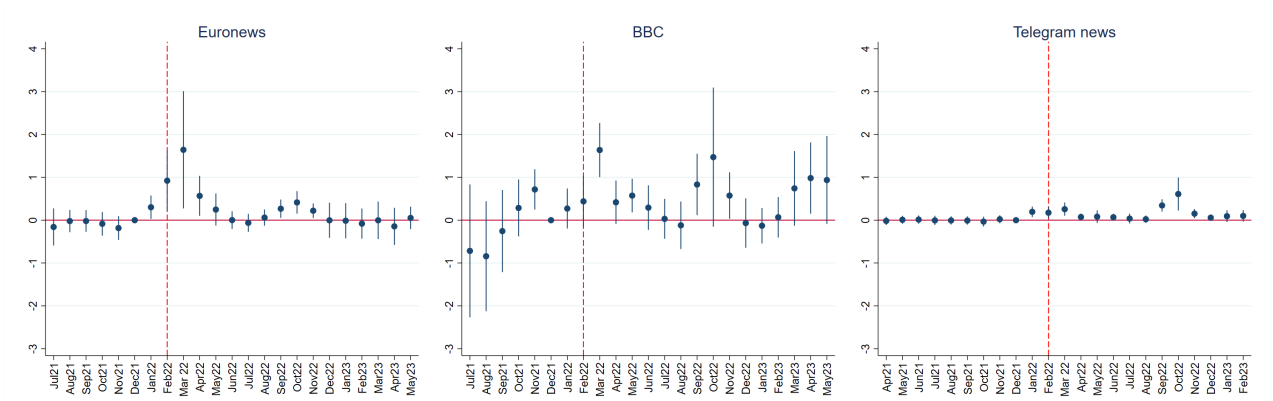
Table D3: Russian language habit and war attitudes

VARIABLES	(1) SMO is justified (binary)	(2)	(3) Russia reponsible	(4)	(5) Ukraine reponsible	(6)	(7) West reponsible	(8)
Russian at home	0.11*** (0.015)	0.090** (0.037)	-0.057*** (0.017)	-0.025 (0.042)	0.013 (0.014)	-0.028 (0.033)	0.087*** (0.012)	0.079*** (0.031)
Constant	0.33*** (0.047)	0.37*** (0.11)	0.20*** (0.049)	0.12 (0.13)	0.21*** (0.044)	0.32*** (0.11)	0.094*** (0.036)	0.041 (0.099)
Observations	6,020	955	6,020	955	6,020	955	6,020	955
R-squared	0.140	0.154	0.084	0.084	0.038	0.067	0.099	0.085
Time FE	Y	Y	Y	Y	Y	Y	Y	Y
Region FE	Y	Y	Y	Y	Y	Y	Y	Y
Mean DV	.257	.248	.262	.293	.201	.179	.114	.139
Sample	All	Managers	All	Managers	All	Managers	All	Managers

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The table shows the regression results (15) for a binary dependent variable “SMO is justified” and the set of binary variables that assign responsibility for the war to Russia, Ukraine, or the West. Robust standard errors are in parentheses.

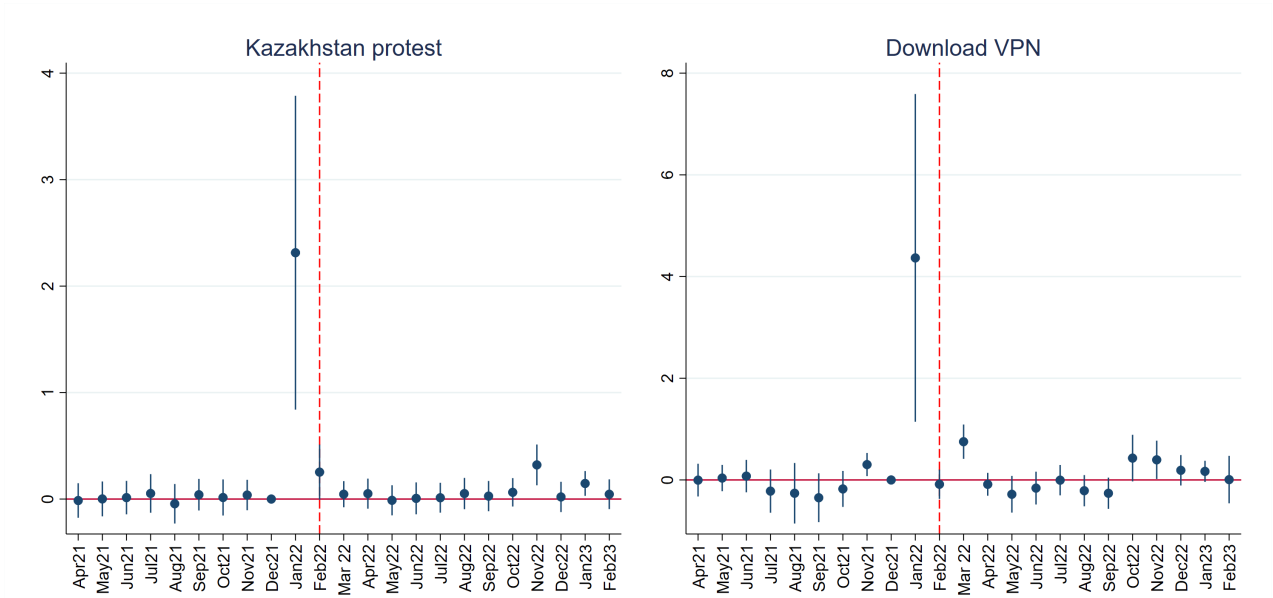
E Appendix

Figure E1: Event-study coefficients for Yandex search of Western outlets and Telegram news



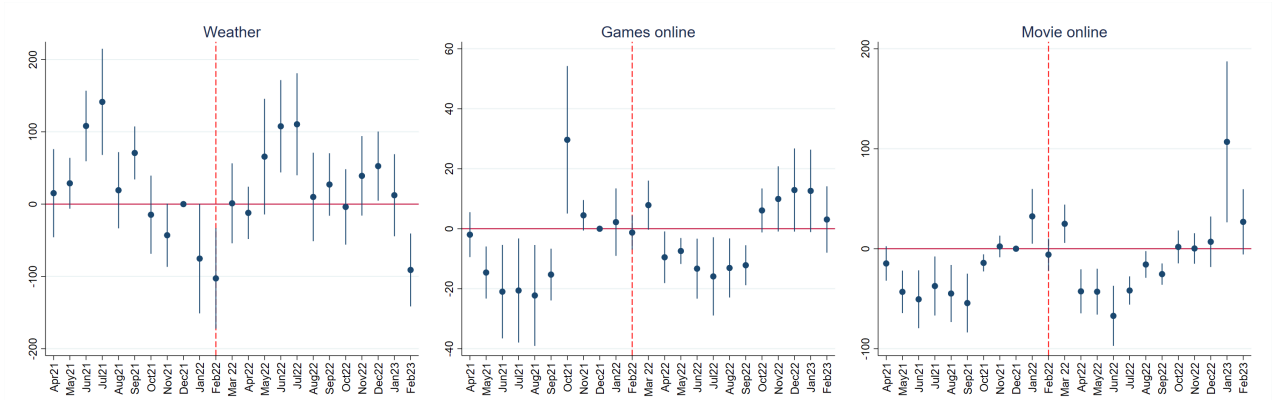
Note. The figure shows coefficients γ_τ from (14) over time. The treatment exposure is the regional share of Russian-speaking Kazakhs. The outcome variable is the number of keyword searches of a keyword in Yandex per 1,000 regional inhabitants.

Figure E2: Event-study coefficients for Yandex searches on the January 2022 protests in Kazakhstan



Note. The figure shows coefficients γ_τ from (14) over time. The treatment exposure is the regional share of Russian-speaking Kazakhs.

Figure E3: Event-study coefficients for Yandex search of weather and entertainment



Note. The figure shows coefficients γ_τ from (14) over time. The treatment exposure is the regional share of Russian-speaking Kazakhs. The outcome variable is the number of keyword searches of a keyword in Yandex per 1,000 regional inhabitants.